Mapping Class Groups: Geometry, Moduli, and Operads

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Plan

1. Mapping Class Groups: Classical Foundations

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[2] Nikolai V. Ivanov, *Mapping class groups*, 2002.

[3] Shigeyuki Morita, *Introduction to mapping class groups of surfaces and related groups*, 2007.

2. Mapping Class Groups: Teichmüller Theory and Geometry

[1] John Hamal Hubbard, *Teichmüller theory and applications to geometry, topology, and dynamics. Vol. 1*, 2006.

[2] Yôichi Imayoshi and Masahiko Taniguchi, *An introduction to Teichmüller spaces*, 1992.

[3] Dimitri Zvonkine, *An introduction to moduli spaces of curves and their intersection theory*, 2012.

3. Mapping Class Groups: Stable Homotopy and Harer Stability

[1] Nathalie Wahl, *The Mumford conjecture, Madsen-Weiss and homological stability for mapping class groups of surfaces*, 2013.

[2] Søren Galatius, *Lectures on the Madsen-Weiss theorem*, 2013.

[3] Allen Hatcher, *A short exposition of the Madsen-Weiss theorem*, 2014.

4. Mapping Class Groups: Operads and Gluing Structures

[1] Olivia Borghi and Marcy Robertson, *Lecture notes on modular infinity operads and Grothendieck-Teichmüller theory*, 2025.

[2] Florian Kranhold, *Coloured topological operads and moduli spaces of surfaces with multiple boundary curves*, 2022.

Mapping Class Groups: Classical Foundations

Topological Group Preliminaries

$\boxed{1}$ **Theorem.** [Sin2019] Let X be a locally connected, locally compact Hausdorff space. Let $\mathrm{Homeo}(X)$ denote the group of all self-homeomorphisms of X, with the composition of mappings as the group operation. Equip $\mathrm{Homeo}(X)$ with the compact-open topology. Then the multiplication and inversion operations in $\mathrm{Homeo}(X)$ are continuous.

$\boxed{2}$ **Theorem.** [Sin2019] Let G be a topological group, and let G_0 denote the path component of the identity element of G. Then G_0 is a normal subgroup of G, and the factor group G/G_0 with the quotient topology is a topological group. Moreover, the cosets of G_0 in G are precisely the set $\pi_0(G)$ of path components of G.

Mapping Class Groups of Surfaces

$\boxed{3}$ **Convention.** All manifolds are assumed Hausdorff and second countable. Unless stated otherwise, S denotes a connected, oriented topological 2-manifold, possibly non-compact and possibly with boundary.

Recall that the group of orientation-preserving homeomorphisms of S that restrict to the identity on ∂S, denoted $\mathrm{Homeo}^+(S, \partial S)$, is a topological group.

$\boxed{4}$ **Definition.** [Deh[1987], Breslau Mathematics Colloquium, 1922] The *mapping class group* of S, denoted $\mathrm{Mod}(S)$, is the factor group $\mathrm{Homeo}^+(S, \partial S) \,/\, \mathrm{Homeo}^+(S, \partial S)_0$.

$\boxed{5}$ **Remark.** $\mathrm{Mod}(S)$ is precisely the set $\pi_0\left(\mathrm{Homeo}^+(S, \partial S)\right)$. Thus, $f, g \in \mathrm{Homeo}^+(S, \partial S)$ represent the same mapping class if and only if there exists an *isotopy* (i.e., a homotopy through homeomorphisms) rel ∂S from f to g.

$\boxed{6}$ **Remark.** $\mathrm{Homeo}^+(S, \partial S)_0 = \mathrm{Homeo}(S, \partial S)_0$, since an isotopy is a proper homotopy through maps $(S, \partial S) \to (S, \partial S)$ [Cro[1963]] and therefore preserves orientation.

Homotopy vs. Isotopy on Surfaces

$\boxed{7}$ **Theorem.** [Bég+2020] [Bae1927; Bae1928; Eps1966] Let $h \in \mathrm{Homeo}^+(S, \partial S)$. Then h is homotopic rel ∂S to the identity if and only if h is isotopic rel ∂S to the identity.

$\boxed{8}$ **Corollary.** Let $\sim$ be the equivalence relation on $\mathrm{Homeo}^+(S, \partial S)$ defined by $f \sim g$ if f is homotopic rel ∂S to g. Then the canonical map from $\mathrm{Mod}(S)$ to the quotient group $\mathrm{Homeo}^+(S, \partial S)/\sim$ is a group isomorphism.

$\boxed{9}$ **Remark.** In [Fri+1986], Friedman–Witt constructed an example of a self-homeomorphism (a sphere twist) of a closed, orientable 3-manifold that is homotopic to the identity but not isotopic to the identity.

$\boxed{10}$ **Theorem.** [Hat[2026]] [Rad[1925]] S admits a smooth structure. Moreover, this smooth structure can be chosen to agree with a given smooth structure on a specified open subset of S.

$\boxed{11}$ **Theorem.** [Hat[2026]] [Bae[1928]; Eps[1966]] Let S be equipped with a smooth structure. Then every self-homeomorphism of S is isotopic to a diffeomorphism. Furthermore, if a homeomorphism is already a diffeomorphism near ∂S, then the isotopy can be chosen to be fixed in a neighborhood of ∂S.

$\boxed{12}$ **Theorem.** [Ban[1997]] Let M be an oriented smooth manifold. Denote by $\mathrm{Diff}^+(M, \partial M)$ the group of orientation-preserving diffeomorphisms of M that restrict to the identity on ∂M. Equipped with the weak C^∞ topology, $\mathrm{Diff}^+(M, \partial M)$ is a topological group.

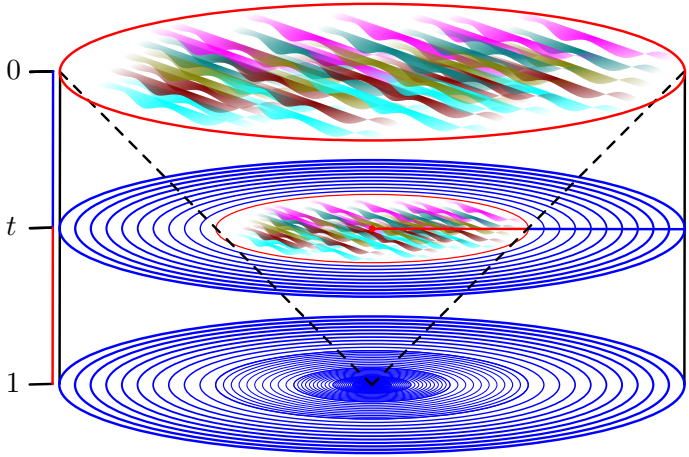
Smooth Mapping Class Groups

$\boxed{13}$ **Theorem.** [Pal+2024] The inclusion $\mathrm{Diff}^+(S, \partial S) \longrightarrow \mathrm{Homeo}^+(S, \partial S)$ is a weak homotopy equivalence of topological groups. In particular, the inclusion induces a bijection $\pi_0(\mathrm{Diff}^+(S, \partial S)) \longrightarrow \mathrm{Mod}(S)$.

$\boxed{14}$ **Theorem.** (Smale Conjecture) [Cer1961; Hat1983] For a compact 3-manifold M, the inclusion $\mathrm{Diff}^+(M, \partial M) \longrightarrow \mathrm{Homeo}^+(M, \partial M)$ is a weak homotopy equivalence.

$\boxed{15}$ **Remark.** In [Rub1998], Ruberman constructed a self-diffeomorphism of a closed, simply connected, smooth 4-manifold X that represents a non-trivial element in the kernel of the inclusion-induced map $\pi_0(\mathrm{Diff}^+(X)) \longrightarrow \pi_0(\mathrm{Homeo}^+(X))$.

16 Theorem. [Ale1924] $\mathrm{Mod}(\mathbb{D})$ is trivial. More generally, $\mathrm{Homeo}^+(\mathbb{D}, \partial\mathbb{D})$ is contractible.



Proof. For $\varphi \in \mathrm{Homeo}^+(\mathbb{D}, \partial\mathbb{D})$, the maps $H_t \colon \mathbb{D} \to \mathbb{D}$ given by $H_t(z) = (1-t)\varphi\left(\frac{z}{1-t}\right)$ for $0 \leq |z| \leq 1-t$ and $H_t(z) = z$ for $1-t \leq |z| \leq 1$ define an isotopy rel $\mathbb{S}^1$ from φ to $\mathrm{id}_{\mathbb{D}}$. $\square$

17 Corollary. $\mathrm{Mod}(\mathbb{D} \setminus \{0\})$ is trivial.

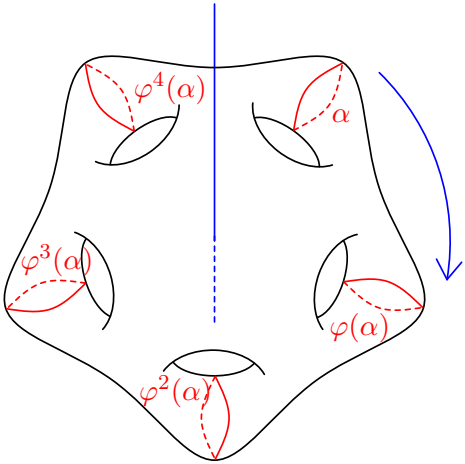
18 Theorem. $\mathrm{Mod}(\mathbb{R}^2)$ is trivial.

19 Theorem. [Deh1922] $\mathrm{Mod}(\mathbb{S}^2)$ is trivial.

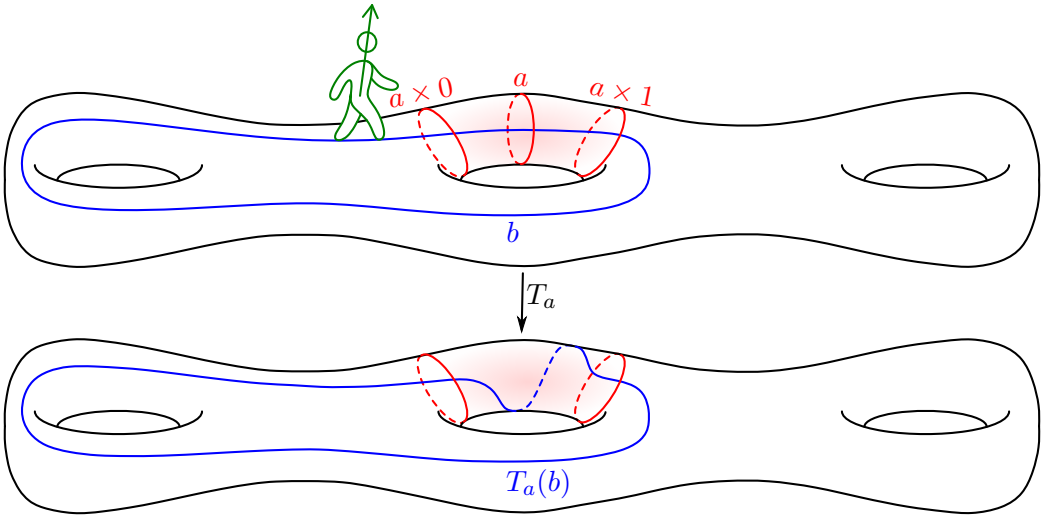
Proof. Let $\varphi \in \mathrm{Homeo}^+(\mathbb{S}^2)$ and $x \in \mathbb{S}^2$. The homogeneity of connected, boundaryless manifolds provides a self-homeomorphism h of $\mathbb{S}^2$ such that $h(\varphi(x)) = x$ and h is homotopic to the identity [Vic1994]. The orientation-preserving self-homeomorphism $h \circ \varphi$ restricts to a homeomorphism of $\mathbb{S}^2 \setminus \{x\} \cong \mathbb{R}^2$, which is homotopic, and hence isotopic, to the identity. This isotopy extends to $\mathbb{S}^2$, showing $h \circ \varphi$ is isotopic to the identity. Therefore, φ is homotopic to the identity. $\square$

20 Example. [Deh1922] Let $S_{g,b,p}$ denote the orientable surface obtained by removing p points from the interior of a compact surface of genus g with b boundary components. Then $\mathrm{Mod}(S_{0,2,0}) \cong \mathbb{Z}$, $\mathrm{Mod}(S_{0,0,3}) \cong \mathrm{Sym}_3$, $\mathrm{Mod}(S_{0,0,2}) \cong \mathbb{Z}/2\mathbb{Z}$, and $\mathrm{Mod}(S_{1,0,0}) \cong \mathrm{SL}(2, \mathbb{Z})$.

21 Lemma. [Eps[1966]] [Bae[1928]] Let C_1 and C_2 be disjoint, homotopic, non-contractible circles embedded in S. Then there exists an annulus A embedded in S such that $\partial A = C_1 \cup C_2$.



Rotation by $2\pi/5$ gives an order 5 element of $\mathrm{Mod}(S_5)$



The *right Dehn twist* T_a acts on $a \times [0,1] \equiv \mathbb{S}^1 \times [0,1]$ by $T_a\left(e^{i\theta}, t\right) = \left(e^{i(\theta+2\pi t)}, t\right)$, and is the identity on $S \setminus (a \times [0,1])$.

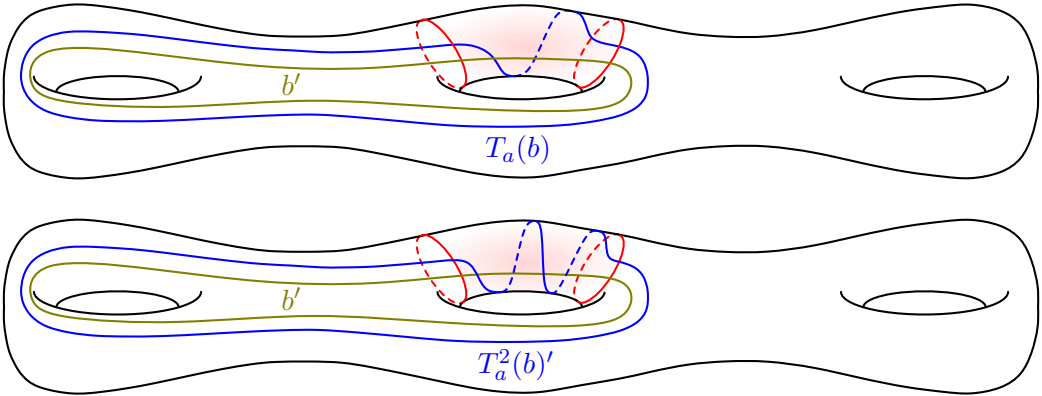
$\boxed{22}$ **Remark.** [Fri2023] The isotopy class of T_a does not depend on the choice of an orientation preserving embedding of the annulus $a \times [0,1]$ in S. Moreover, if a' is another embedded circle in S homotopic to a through embedded circles, then T_a is isotopic to $T_{a'}$.

$\boxed{23}$ **Definition.** Let a and b be embedded circles in S. Denote their free homotopy classes by $\widehat{a}$ and $\widehat{b}$, respectively. The *geometric intersection number* between $\widehat{a}$ and $\widehat{b}$ is

$$i\left(\widehat{a}, \widehat{b}\right) := \min\left\{ |a' \cap b'| : a' \in \widehat{a},\ b' \in \widehat{b} \right\}.$$

$\boxed{24}$ **Theorem.** [Far+2012] Let a and b be embedded non-contractible circles in S such that neither a nor b can be homotoped into ∂S or into a closed subset A of S with $A \cong \mathbb{D} \setminus \{0\}$. Then

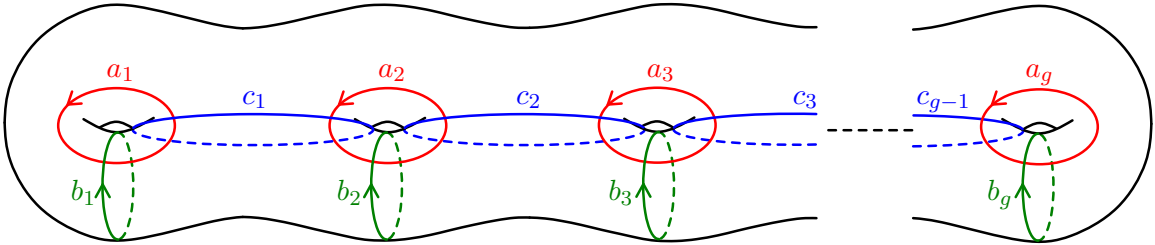
$$i\left(\widehat{T_a^k(b)}, \widehat{b}\right) = |k| \cdot i\left(\widehat{a}, \widehat{b}\right) \quad \text{for every } k \in \mathbb{Z}.$$



$$b' \cap T_a(b) \;=\; 1 \;=\; i\left(\widehat{T_a(b)}, \widehat{b}\right) \qquad \text{and} \qquad b' \cap T_a(b)' \;=\; 2 \;=\; i\left(\widehat{T_a^2(b)}, \widehat{b}\right)$$

Dehn–Lickorish–Humphries Generators

25 Theorem. [Deh[1938]] [Lic[1964]] For $g \geq 1$, the Dehn twists about the circles $a_1, \ldots, a_g$, $b_1, \ldots, b_g$, $c_1, \ldots, c_{g-1}$ generate $\mathrm{Mod}(S_g)$.



26 Theorem. [Hum[1979]] For any $g \geq 1$, the Dehn twists about the circles $a_1, \ldots, a_g$, $c_1, \ldots, c_{g-1}$, b_{j-1}, b_j, fo any $2 \leq j \leq g$, generate $\mathrm{Mod}(S_g)$. Moreover, any set of Dehn twist generators for $\mathrm{Mod}(S_g)$ must contain at least $2g+1$ elements.

Symplectic Representation of Mapping Class Group

$\boxed{27}$ **Aim.** Describe the image of the natural representation $\Psi\colon \operatorname{Mod}(S_g) \to \operatorname{Aut}(H_1(S_g))$ induced by the action of $\operatorname{Homeo}^+(S_g)$ on $H_1(S_g)$.

Define the *intersection form* $\omega\colon H_1(S_g) \times H_1(S_g) \to \mathbb{Z}$ by

$$\omega(v, w) := (\operatorname{PD}(v) \smile \operatorname{PD}(w)) \frown [S_g]\,.$$

By choosing a *symplectic basis* for $(H_1(S_g), \omega)$, for instance $([a_1], [b_1], \ldots, [a_g], [b_g])$, we obtain an isomorphism between the following groups:

- $\operatorname{Sp}(H_1(S_g), \omega) := \{\phi \in \operatorname{Aut}(H_1(S_g)) : \omega(\phi(v), \phi(w)) = \omega(v, w) \text{ for all } v, w \in H_1(S_g)\}$
- $\operatorname{Sp}(2g, \mathbb{Z}) \qquad := \left\{M \in \operatorname{GL}(2g; \mathbb{Z}) : M^T \cdot \begin{pmatrix} 0 & \mathrm{I}_g \\ -\mathrm{I}_g & 0 \end{pmatrix} \cdot M = \begin{pmatrix} 0 & \mathrm{I}_g \\ -\mathrm{I}_g & 0 \end{pmatrix}\right\}.$

The image of Ψ is contained in $\operatorname{Sp}(H_1(S_g), \omega)$. The group $\operatorname{Sp}(2g, \mathbb{Z})$ is called the *Siegel modular group of degree g*.

$\boxed{28}$ **Theorem.** [Far+2012] [Put2018] [Mee+1978] [Bur1889] The image of Ψ is equal ($\cong$) to $\operatorname{Sp}(2g, \mathbb{Z})$.

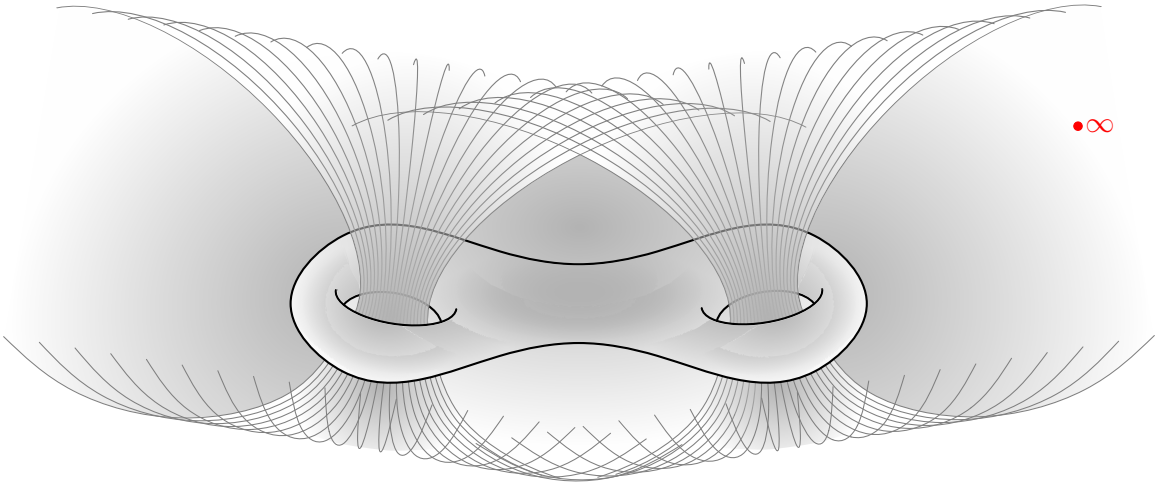
Heegaard Splittings of 3-Manifolds

$\boxed{29}$ **Definition.** A *genus g handlebody* is any compact, connected, orientable 3-manifold that is diffeomorphic to the compact region in $\mathbb{R}^3$ bounded by a standardly embedded S_g.

$\boxed{30}$ **Definition.** Let M be a closed, connected, oriented 3-manifold. A *Heegaard splitting of M of genus g* consists of two smooth submanifolds X and Y of M such that both X and Y are genus g handlebodies, $M = X \cup Y$, and $\partial X = \partial Y = X \cap Y$.

$\boxed{31}$ **Theorem.** [Hee[1898]] Every closed, connected, orientable 3-manifold admits a Heegaard splitting.

$\boxed{32}$ **Remark.** [Fri[2023]] Let $\mathcal{H}$ and $\mathcal{H}'$ be oriented genus g handlebodies, and let $\varphi\colon \partial\mathcal{H} \to \partial\mathcal{H}'$ be an orientation-reversing diffeomorphism. Then the images of $\mathcal{H}$ and $\mathcal{H}'$ in the oriented manifold $\mathcal{H} \cup_\varphi \mathcal{H}' := \mathcal{H} \sqcup \mathcal{H}' \,/\, p \sim \varphi(p)$ form a Heegaard splitting of genus g.



A Heegaard splitting of $\mathbb{S}^3$ of genus 2

Homology 3-Spheres via Torelli Groups

$\boxed{33}$ **Definition.** The kernel of the representation $\Psi\colon \operatorname{Mod}(S_g) \to \operatorname{Aut}(H_1(S_g))$, denoted $\mathcal{I}(S_g)$, is called the *Torelli subgroup* of $\operatorname{Mod}(S_g)$.

$\boxed{34}$ **Example.** The Dehn twist about a circle c lies in $\mathcal{I}(S_g)$ if and only if c separates S_g.

$\boxed{35}$ **Proposition.** Fix a Heegaard splitting $\mathbb{S}^3 \cong \mathcal{H} \cup_\varphi \mathcal{H}'$ of $\mathbb{S}^3$, and let $f \in \operatorname{Homeo}^+(\partial\mathcal{H})$. Define

$$M_f := \mathcal{H} \cup_{\varphi\circ f} \mathcal{H}'.$$

Then M_f is an oriented 3-manifold. Moreover, we have the following:

- If f is isotopic to f', then there exists an orientation-preserving homeomorphism between M_f and $M_{f'}$.
- If f represents an element of $\mathcal{I}(\partial\mathcal{H})$, then M_f is an integral homology 3-sphere.

$\boxed{36}$ **Theorem.** [Mor[1989]] Every integral homology 3-sphere is homeomorphic to some M_f.

Artin Braids and Configuration Spaces

$\boxed{37}$ **Motivation.** Fix an n-point set $\mathscr{P} = \{p_1, \ldots, p_n\} \subset \mathbb{D} \setminus \partial\mathbb{D}$ and let $\varphi \in \mathrm{Homeo}^+(\mathbb{D}, \partial\mathbb{D})$ such that $\varphi(\mathscr{P}) = \mathscr{P}$. By Alexander's trick, there exists a path $\{\varphi_t\}_{t\in[0,1]}$ in $\mathrm{Homeo}^+(\mathbb{D}, \partial\mathbb{D})$ from $\mathrm{id}_{\mathbb{D}}$ to φ. The n-tuple of paths $b_k \colon [0,1] \to \mathbb{C}$ defined by $b_k(t) \coloneqq \varphi_t(p_k)$ then satisfies the following three properties:

(1) $b_k(0) = p_k$ for all $1 \leq k \leq n$;

(2) there exists a permutation $\chi = \theta(\beta) \in \mathrm{Sym}_n$ such that $b_k(1) = p_{\chi(k)}$ for all $1 \leq k \leq n$;

(3) $b_k(t) \neq b_l(t)$ for all $k \neq l$ and all $t \in [0,1]$.

$\boxed{38}$ **Definition.** [Par[2009]] Fix an n-point set $\mathscr{P} = \{p_1, \ldots, p_n\} \subset \mathbb{C}$. A *braid on n strands* is an n-tuple $\beta = (b_1, \ldots, b_n)$ of paths $b_k \colon [0,1] \to \mathbb{C}$ such that such that properties (1)–(3) hold.

$\boxed{39}$ **Note.** Unless otherwise stated, we assume $p_k = k$ for all $1 \leq k \leq n$.

$\boxed{40}$ **Definition.** Let M be a smooth manifold. The *nth ordered configuration space of M* is the subspace of M^n given by: $\mathsf{C}_M(n) := \{(z_1, \ldots, z_n) \in M^n \mid z_i \neq z_j \text{ for all } i \neq j\}$. The symmetric group Sym_n acts freely on $\mathsf{C}_M(n)$ by permuting the coordinates. The *nth unordered configuration space of M* is given by the orbit space $\mathsf{UC}_M(n) := \mathsf{C}_M(n)/\mathrm{Sym}_n$.

The braid β can be thought of as a based loop $\ell_\beta \colon (\mathbb{S}^1, 1) \to (\mathsf{UC}_\mathbb{C}(n), \mathscr{P})$. If β' is another braid, then we say β and β' are *homotopic* if their corresponding based loops ℓ_β and $\ell_{\beta'}$ are homotopic through based loops $(\mathbb{S}^1, 1) \to (\mathsf{UC}_\mathbb{C}(n), \mathscr{P})$.

$\boxed{41}$ **Remark.** The quotient map $q \colon \mathsf{C}_\mathbb{C}(n) \to \mathsf{UC}_\mathbb{C}(n)$ is a regular $n!$-sheeted covering map. Hence, if β and β' are homotopic braids, the homotopy lifting property implies that $\theta(\beta) = \theta(\beta')$.

$\boxed{42}$ **Remark.** Forgetting the last coordinate gives a fiber bundle $\mathsf{C}_\mathbb{C}(n+1) \to \mathsf{C}_\mathbb{C}(n)$ with fiber over $(1, \ldots, n)$ equal to $\mathbb{C} \setminus \{1, \ldots, n\}$ [Fad+1962]. Hence the finite-dimensional CW complexes $\mathsf{C}_\mathbb{C}(n)$ and $\mathsf{UC}_\mathbb{C}(n)$ are aspherical, so their fundamental groups are torsion-free.

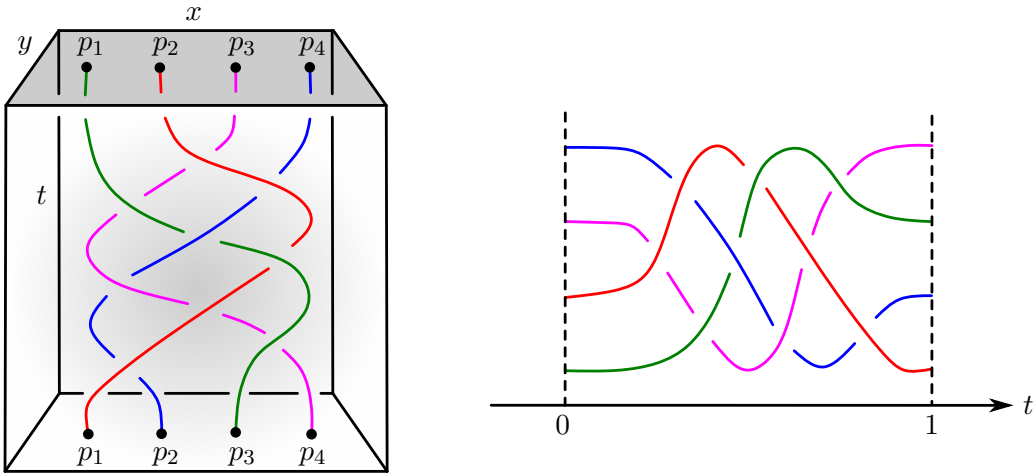
Braid Diagrams and Braid Groups

$\boxed{43}$ **Definition.** [Par2009] For $1 \leq k \leq n$, let I_k be a copy of the interval $[0,1]$. Take a braid $\beta = (b_1, \ldots, b_n)$ and define the *geometric braid*

$$\beta^g \colon I_1 \sqcup \cdots \sqcup I_n \to \mathbb{R}^2 \times [0,1] \text{ by } \beta^g(t) = (b_k(t), t) \text{ for all } t \in I_k \text{ and all } 1 \leq k \leq n.$$

Let $\mathrm{proj} \colon \mathbb{R}^2 \times [0,1] \to \mathbb{R} \times [0,1]$ be the projection defined by $\mathrm{proj}(x,y,t) = (x,t)$.

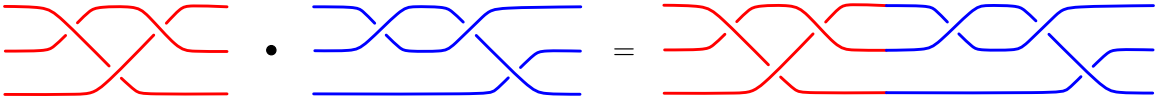
Up to homotopy, we can assume that $\mathrm{proj} \circ \beta^g$ is a smooth immersion with only transversal double points, which we call *crossings*. In each crossing we indicate graphically which strand goes over the other. Such a representation of β is called a *braid diagram* of β.



A geometric braid on 4 strands and its diagram

44 Definition. The *product of two braids* $\alpha = (a_1, \ldots, a_n)$ and $\beta = (b_1, \ldots, b_n)$ is defined via path concatenation:

$$\alpha \cdot \beta = \left(a_1 b_{\chi(1)}, \ldots, a_n b_{\chi(n)}\right), \text{ where } \chi = \theta(\alpha).$$



Product of two braids

45 Theorem. [Par2009] Let $\mathcal{B}_n$ denote the set of homotopy classes of braids on n strands. Then the product of braids defined above induces a well-defined group operation on $\mathcal{B}_n$.

- The identity of this group is represented by the trivial braid $\mathcal{C} = (\mathcal{C}_1, \ldots, \mathcal{C}_n)$, where, for $1 \leq k \leq n$, $\mathcal{C}_k$ denotes the constant path at p_k.
- The inverse of a given braid class is obtained by taking the mirror image of a representative (geometric) braid across the horizontal mid-plane.

Surface Braid Groups and Mapping Class Groups

$\boxed{46}$ **Definition.** Fix an n-point set $\mathscr{P} = \{p_1, \ldots, p_n\} \subset S \setminus \partial S$. Let $\mathrm{Mod}(S, \mathscr{P})$ denote the set of all path components of $\mathrm{Homeo}^+(S, \partial S, \mathscr{P}) := \{\, \varphi \in \mathrm{Homeo}^+(S, \partial S) \mid \varphi(\mathscr{P}) = \mathscr{P} \,\}$.

Replacing $\mathbb{C}$ by S in the definition of braids, one may define a *braid of S based at $\mathscr{P}$*. The homotopy classes of braids on S based at $\mathscr{P}$ form a group, denoted $\mathcal{B}_n(S, \mathscr{P})$, called the *braid group of S on n strands based at $\mathscr{P}$*. We may think of $\mathcal{B}_n(S, \mathscr{P})$ as the fundamental group of $\mathsf{UC}_S(n)$ based at $\mathscr{P}$.

$\boxed{47}$ **Theorem.** [Bir[1969]] Suppose S is compact and not homeomorphic to either the sphere or the torus. Then we have the short exact sequence

$$1 \longrightarrow \mathcal{B}_n(S, \mathscr{P}) \longrightarrow \mathrm{Mod}(S, \mathscr{P}) \longrightarrow \mathrm{Mod}(S) \longrightarrow 1.$$

Proof. The map

$$\mathrm{Homeo}^+(S, \partial S) \longrightarrow \mathsf{UC}_S(n) \qquad \varphi \longmapsto \{\varphi(p_1), \ldots, \varphi(p_n)\}$$

is a fiber bundle whose fiber over $\mathscr{P}$ being $\mathrm{Homeo}^+(S, \partial S, \mathscr{P})$. Furthermore, every path component of $\mathrm{Homeo}^+(S, \partial S)$ is contractible [Ham[1966]]. Thus, by the homotopy long exact sequence of a fiber bundle, we obtain the short exact sequence

$$1 \longrightarrow \pi_1\left(\mathsf{UC}_S(n), \mathscr{P}\right) \longrightarrow \pi_0\left(\mathrm{Homeo}^+(S, \partial S, \mathscr{P})\right) \longrightarrow \pi_0\left(\mathrm{Homeo}^+(S, \partial S)\right) \longrightarrow 1 \quad \square$$

$\boxed{48}$ **Corollary.** (Birman exact sequence) Suppose S is compact and not homeomorphic to either the sphere or the torus. If $x \in S \setminus \partial S$, then we have a short exact sequence

$$1 \longrightarrow \pi_1(S, x) \longrightarrow \mathrm{Mod}(S \setminus \{x\}) \longrightarrow \mathrm{Mod}(S) \longrightarrow 1.$$

$\boxed{49}$ **Theorem.** [Art[1925]; Art[1947]] $\mathrm{Mod}(\mathbb{D} \setminus \mathscr{P}) \equiv \mathrm{Mod}(\mathbb{D}, \mathscr{P}) \cong \mathcal{B}_n(\mathbb{D}, \mathscr{P}) \equiv \mathcal{B}_n$.

Motivating the Dehn-Nielsen-Baer Theorem

- Earlier, we studied the action of $\mathrm{Mod}(S_g)$ on $H_1(S_g)$ and showed that the symplectic representation $\mathrm{Mod}(S_g) \to \mathrm{Aut}(H_1(S_g))$ naturally surjects onto $\mathrm{Sp}(2g; \mathbb{Z})$.

- The difficulty in defining an action of $\mathrm{Mod}(S_g)$ on $\pi_1(S_g, *)$ lies in the fact that neither an arbitrary homeomorphism nor an isotopy preserves $*$.

- Since surfaces of genus $g \geq 1$ are aspherical, we can bypass this ambiguity. The failure to preserve the basepoint simply translates to an inner automorphism of $\pi_1(S_g, *)$.

Let X be a path connected CW-complex with basepoint $* \in X$, and let $\pi = \pi_1(X, *)$. Define

$$\langle X, X \rangle := \{F \colon X \to X \mid F(*) = *\}/\simeq \quad \text{and} \quad [X, X] := \{f \colon X \to X\}/\sim$$

where $\simeq$ is homotopy through basepoint-preserving maps, and $\sim$ is free homotopy.

$\boxed{50}$ **Remark.** The forgetful map $\langle X, X \rangle \to [X, X]$ is surjective: Given a map f and a path γ from $*$ to $f(*)$, the homotopy extension property yields a homotopy H from f to a based map F. Note that the basepoint-change isomorphism $\gamma_\sharp$ along $\gamma := H(*, -)$ satisfies

$$F_* = \gamma_\sharp \circ f_*.$$

$\boxed{51}$ **Remark.** If H is a free homotopy from a based map F to another based map G, then

$$F_* = c_y \circ G_*,$$

where y is the homotopy class of the loop $H(*, -)$ in π, and c_y is the inner automorphism of π defined by $c_y(z) := yzy^{-1}$.

$\boxed{52}$ **Note.** If $f\colon X \to X$ is a homotopy equivalence, and μ and ν are paths from $*$ to $f(*)$, then the automorphisms $\nu_\sharp \circ f_*$ and $\mu_\sharp \circ f_*$ determine the same right coset in the outer automorphism group

$$\mathrm{Out}(\pi) := \mathrm{Aut}(\pi)/\,\mathrm{Inn}(\pi).$$

Indeed, $\nu_\sharp \circ f_* = (\nu * \overline{\mu})_\sharp \circ \mu_\sharp \circ f_* = c_{[\nu * \overline{\mu}]} \circ (\mu_\sharp \circ f_*)$.

$\boxed{53}$ **Theorem.** [Whi[1978]] If X is also aspherical, then the map

$$\langle X, X\rangle \longrightarrow \mathrm{Hom}(\pi,\pi), \quad [F] \longmapsto F_*,$$

is a bijection.

$\boxed{54}$ **Theorem.** [Whi[1978]] If X is also aspherical, and F and G are based maps such that

$$F_* = c_y \circ G_*$$

for some $y \in \pi$, then F is freely homotopic to G.

Dehn-Nielsen-Baer-Epstein Theorem

$\boxed{55}$ **Theorem.** [Bae[1928]; Eps[1966]] Suppose S is aspherical and boundaryless with basepoint $* \in S$, and let $\pi = \pi_1(S, *)$. Then the map

$$\sigma_g \colon \operatorname{Mod}(S) \longrightarrow \operatorname{Out}(\pi), \qquad [\varphi] \longmapsto \nu_\sharp \circ \varphi_* \cdot \operatorname{Inn}(\pi)$$

is a well-defined injective group homomorphism, where $\varphi \in \operatorname{Homeo}^+(S)$ and $\nu \colon [0,1] \to S$ is a path from $*$ to $\varphi(*)$.

Mapping Class Groups: Teichmüller Theory and Geometry

Target: Parameterize the Complex Structures on a Surface of Given Genus

§ Riemann Surfaces: Definition and Examples.

$\boxed{56}$ **Definition.** Let X be a Hausdorff space. A *complex atlas of dimension* 1 on X is a family of functions $\{\varphi_i \colon U_i \to \mathbb{C}\}_{i \in \mathcal{I}}$, called *charts*, satisfying the following conditions:

- U_i is an open subset of X for each $i \in \mathcal{I}$.
- The chart φ_i maps U_i homeomorphically onto an open subset of $\mathbb{C}$ for each $i \in \mathcal{I}$.
- The collection $\{U_i\}_{i \in \mathcal{I}}$ is a cover of X.
- If $U_i \cap U_j \neq \varnothing$, then the transition function

$$\Phi_{ij} := \varphi_j \circ \varphi_i^{-1} \colon \varphi_i(U_i \cap U_j) \to \varphi_j(U_i \cap U_j)$$

  is holomorphic (the inverse of Φ_{ij} is also holomorphic by a result in complex analysis).

$\boxed{57}$ **Definition.** A *Riemann surface* is a pair $(X, \mathfrak{U})$ consisting of a connected Hausdorff space X equipped with a *1-dimensional complex structure* $\mathfrak{U}$ (that is, a maximal complex atlas of dimension 1).

$\boxed{58}$ **Theorem.** [Rad[1925]] [For[1981]] [Hub[2006]] Every Riemann surface is second-countable.

$\boxed{59}$ **Theorem.** [Cal+[1953]] [Hub[2006]] A connected Hausdorff space equipped with a maximal complex atlas of complex dimension ≥ 2 need not be second-countable.

$\boxed{60}$ **Example.** The complex atlas on $\mathbb{C}$ consisting of the single chart $\mathrm{id}_{\mathbb{C}}$ makes $\mathbb{C}$ a Riemann surface.

$\boxed{61}$ **Example.** Let $\widehat{\mathbb{C}} := \mathbb{C} \sqcup \{\infty\} \cong \mathbb{S}^2$. The complex atlas on $\widehat{\mathbb{C}}$ consisting of the charts

$$\widehat{\mathbb{C}} \setminus \{\infty\} \to \mathbb{C}, \quad z \longmapsto z \qquad \text{and} \qquad \widehat{\mathbb{C}} \setminus \{0\} \to \mathbb{C}, \quad z \longmapsto 1/z$$

makes $\widehat{\mathbb{C}}$ a Riemann surface, called the *Riemann sphere*.

$\boxed{62}$ **Theorem** (Canonical Orientation)**.** Let $(X, \mathfrak{U})$ be a Riemann surface. Then the maximal complex atlas $\mathfrak{U}$ is oriented; that is, the Jacobian determinant of any transition map is everywhere strictly positive. (In general, any complex manifold is canonically oriented.)

Proof. A transition function Φ_{ij} can be written as $\Phi_{ij} = u + iv$, where u and v are real valued functions. Let $z \in \varphi_i(U_i \cap U_j)$. By the Cauchy-Riemann equations ($u_x = v_y$, $u_y = -v_x$), the determinant of the Jacobian matrix of Φ_{ij} at z is equal to

$$u_x(z)v_y(z) - u_y(z)v_x(z) = (u_x(z))^2 + (v_x(z))^2 = |\Phi'_{ij}(z)|^2,$$

which is positive as Φ_{ij} is a biholomorphism. $\square$

$\boxed{63}$ **Definition.** Let X and Y be Riemann surfaces, equipped with maximal complex atlases $\{(U_\alpha, \varphi_\alpha)\}_{\alpha \in \mathscr{A}}$ and $\{(V_\beta, \psi_\beta)\}_{\beta \in \mathscr{B}}$, respectively. A function $f\colon X \to Y$ is called *holomorphic* if

$$\psi_\beta \circ f \circ \varphi_\alpha^{-1}\colon \varphi_\alpha\big(U_\alpha \cap f^{-1}(V_\beta)\big) \to \psi_\beta\big(f(U_\alpha) \cap V_\beta\big)$$

is holomorphic for all $\alpha \in \mathscr{A}$ and $\beta \in \mathscr{B}$ such that $f(U_\alpha) \cap V_\beta \neq \varnothing$.

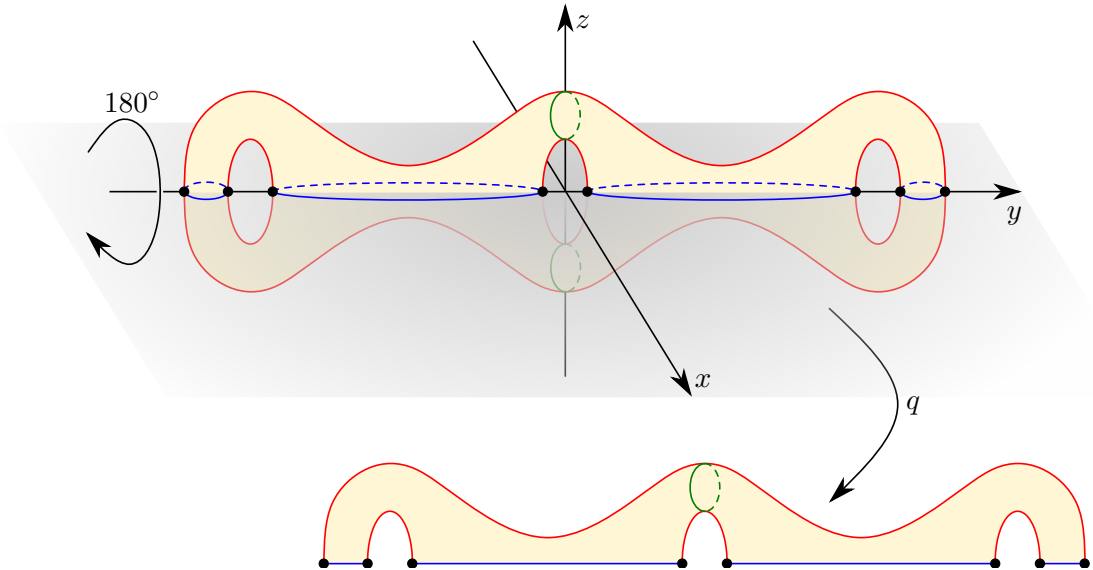
A bijective holomorphic function is called a *biholomorphism*. The *automorphism group* of X, denoted by $\mathrm{Aut}(X)$, is the set of all biholomorphisms from X to itself.

§ Riemann Surfaces from Polynomials: The Hyperelliptic Case.

$\boxed{64}$ **Theorem** (*Pullback Structure* [For[1981]]). If Y is a Riemann surface, X is a connected Hausdorff space, and $p\colon X \to Y$ is a local homeomorphism, then X admits a unique 1-dimensional complex structure such that p is holomorphic.

$\boxed{65}$ **Theorem.** [Ful[1995]] Let Y be a Riemann surface, B a finite subset of Y, and $p\colon X^\circ \to Y \setminus B$ a finite-sheeted covering map, where X° is a connected Hausdorff space.

- **Existence:** There is an embedding of X° as an open subset of a Riemann surface X that is the union of X° and a finite set, such that p extends to a proper holomorphic map $\widetilde{p}\colon X \to Y$.
- **Uniqueness:** If X' is another Riemann surface satisfying the above conditions, with a proper holomorphic map $\widetilde{p}'\colon X' \to Y$ extending p, then there exists a biholomorphism $\Phi\colon X \to X'$ such that $\Phi|_{X^\circ} = \mathrm{id}_{X^\circ}$ and $\widetilde{p}' \circ \Phi = \widetilde{p}$.



The branched covering $q\colon S_3 \to \mathbb{S}^2$ induced by the hyperelliptic involution $(x, y, z) \mapsto (-x, y, -z)$.

$\boxed{66}$ **Example.** Every compact, connected, orientable surface without boundary admits a 1-dimensional complex structure. This can be seen by considering a 2-sheeted covering map $p\colon S_{g,0,2g+2} \longrightarrow S_{0,0,2g+2}$ for any $g \geq 1$.

$\boxed{67}$ **Theorem.** [Gol[1971]] Let S be a connected, non-compact, orientable surface without boundary. For any integer $g \geq 2$, there exists a covering map from S to S_g. Thus, S admits a 1-dimensional complex structure.

$\boxed{68}$ **Theorem.** [Ful[1995]] [Gir+[2012]] Let n be a positive integer, and let

$$F(z,w) = a_0(z)w^n + a_1(z)w^{n-1} + \cdots + a_n(z)$$

be an irreducible polynomial in $\mathbb{C}[z,w]$ with $a_0(z) \neq 0$. Let $d(z)$ be a non-zero polynomial guaranteed by Bézout's identity such that

$$B_1(z,w)F(z,w) + B_2(z,w)F_w(z,w) = d(z) \tag{†}$$

for some polynomials $B_1, B_2 \in \mathbb{C}[z, w]$. Define $S = \{z \in \mathbb{C} \mid a_0(z)d(z) = 0\}$, and let

$$X^\circ := \left\{(z, w) \in \mathbb{C}^2 \mid F(z, w) = 0 \text{ and } z \notin S\right\}.$$

Then the projection map

$$p\colon X^\circ \to \mathbb{C} \setminus S \qquad \text{given by} \qquad p(z, w) = z$$

is a (unbranched) n-sheeted covering, and the space X° is connected.

Consequently, there exists a compact Riemann surface X, called the *Riemann surface of the polynomial F*, and a proper holomorphic mapping $\widetilde{p}\colon X \to \widehat{\mathbb{C}}$ such that X° embeds as an open subset of X with $X \setminus X^\circ$ being a finite set, and $\widetilde{p}|_{X^\circ} = p$.

69 **Theorem.** [Ful[1995]] [Gir+[2012]] Every compact Riemann surface is the Riemann surface of some polynomial.

$\boxed{70}$ **Example.** Let $F(z,w) = aw + bz + c$, where $a \in \mathbb{C} \setminus \{0\}$. Then the Riemann surface of F is homeomorphic to $\mathbb{S}^2$.

$\boxed{71}$ **Example.** Let $g \geq 1$, and let X be the Riemann surface of

$$F(z,w) = w^2 - (z - a_1) \cdots (z - a_{2g+2}) \tag{$*$}$$

where all a_i are distinct complex numbers. A valid choice of polynomials satisfying Bézout's identity (†) is given by:

$$B_1(z,w) = 1, \quad B_2(z,w) = -w/2, \quad \text{and} \quad d(z) = -(z - a_1) \cdots (z - a_{2g+2}).$$

Then $S = \{a_1, \ldots, a_{2g+2}\}$. The 2-sheeted covering $p\colon X^\circ \to \mathbb{C} \setminus S$ extends to a proper holomorphic map $\widetilde{p}\colon X \to \widehat{\mathbb{C}}$, and $X \setminus X^\circ$ consists of exactly $2g+4$ points.

The set of branch points of $\widetilde{p}$ is exactly S, and $\#\widetilde{p}^{-1}(S) = \#S$. By the Riemann-Hurwitz formula

$$\chi(X) - \#\widetilde{p}^{-1}(S) = \deg p \cdot \left(\chi(\widehat{\mathbb{C}}) - \#S\right),$$

we have $\chi(X) = 2 - 2g$. Thus, X is homeomorphic to S_g.

$\boxed{72}$ **Remark.** The self-map $(z, w) \longmapsto (z, -w)$ of X° extends to an automorphism ι of X such that $\iota^2 = \mathrm{id}_X$ and ι has exactly $2g + 2$ fixed points.

$\boxed{73}$ **Definition.** A compact Riemann surface of genus $g \geq 0$ is said to be *hyperelliptic* if it admits an *involution* (an automorphism of order two) that fixes exactly $2g + 2$ points.

$\boxed{74}$ **Theorem.** [Far+1992] [Gir+2012] Let S be a compact Riemann surface of genus g. Then we have the following:

- Every non-identity element of $\mathrm{Aut}(S)$ has at most $2g + 2$ fixed points.
- If $g \geq 2$, then any two involutions of S each having $2g + 2$ fixed points are equal.
- If S is hyperelliptic and $g \geq 1$, then S is the Riemann surface of some polynomial of the form $(*)$.

$\boxed{75}$ **Theorem.** [Gir+2012] The genus g Riemann surfaces of the polynomials

$$F(z,w) = w^2 - (z-a_1)\cdots(z-a_{2g+2}) \qquad \text{and} \qquad G(z,w) = w^2 - (z-b_1)\cdots(z-b_{2g+2})$$

are biholomorphic if and only if there exists an automorphism M of $\widehat{\mathbb{C}}$ such that

$$M\left(\{a_1,\ldots,a_{2g+2}\}\right) = \{b_1,\ldots,b_{2g+2}\}\,.$$

$\boxed{76}$ **Proposition.** Given $\zeta_1,\zeta_2,\zeta_3,\xi_1,\xi_2,\xi_3 \in \widehat{\mathbb{C}}$ with $\zeta_i \neq \zeta_j$ and $\xi_i \neq \xi_j$ for all $i,j \in \{1,2,3\}$ with $i \neq j$, there exists a unique automorphism M of $\widehat{\mathbb{C}}$ such that $M(\zeta_j) = \xi_j$ for $j = 1,2,3$.

$\boxed{77}$ **Intuition:** The complex dimension of the space of biholomorphism classes of compact Riemann surfaces of genus $g \geq 1$ is at least

$$\dim_{\mathbb{C}}\left(\mathsf{UC}_{\widehat{\mathbb{C}}}(2g+2)/\operatorname{Aut}(\widehat{\mathbb{C}})\right) = \dim_{\mathbb{C}}\mathsf{UC}_{\widehat{\mathbb{C}}}(2g+2) - \dim_{\mathbb{C}}\operatorname{Aut}(\widehat{\mathbb{C}}) = (2g+2)-3 = 2g-1.$$

$\boxed{78}$ **Theorem.** [Far+1992] Every compact Riemann surface of genus ≤ 2 is hyperelliptic. In contrast, for genus greater than 2, not every compact Riemann surface is hyperelliptic.

§ Riemann Uniformization Theorem.

$\boxed{79}$ **Theorem.** [Koe1907[1]] [Koe1907[2]] [Poi1907] A simply connected Riemann surface is biholomorphic to exactly one of the following: $\widehat{\mathbb{C}}$, $\mathbb{C}$, or $\mathbb{H}^2 := \{z \in \mathbb{C} : \mathrm{Im}(z) > 0\}$.

$\boxed{80}$ **Theorem.** Every automorphism of $\mathbb{C}$ is of the form $z \longmapsto az + b$ for some $a \in \mathbb{C} \setminus \{0\}$ and $b \in \mathbb{C}$.

$\boxed{81}$ **Theorem.** The automorphism group of $\widehat{\mathbb{C}}$ can be identified with $\mathrm{PSL}(2, \mathbb{C})$, where the action of a representative matrix $\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$ on $z \in \widehat{\mathbb{C}}$ is described as follows:

- If $z \neq \infty$, then $\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \cdot z := \begin{cases} \frac{az+b}{cz+d} & \text{if } cz + d \neq 0, \\ \infty & \text{if } cz + d = 0, \end{cases}$
- If $z = \infty$, then $\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \cdot \infty := \begin{cases} \frac{a}{c} & \text{if } c \neq 0, \\ \infty & \text{if } c = 0. \end{cases}$

$\boxed{82}$ **Theorem.** The automorphism group of $\mathbb{H}^2$ can be identified with $\mathrm{PSL}(2,\mathbb{R})$, where the action of a representative matrix $\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$ on $z \in \mathbb{H}^2$ is given by $\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \cdot z := \frac{az+b}{cz+d}$.

$\boxed{83}$ **Theorem.** [Nap+2011] Up to biholomorphism, every Riemann surface is biholomorphic to exactly one of the following:

1. $\widehat{\mathbb{C}}$;
2. $\mathbb{C}$;
3. $\mathbb{C}/\Lambda$, where $\Lambda = \mathbb{Z}\xi$ with $\xi \in \mathbb{C} \setminus \{0\}$ acting on $\mathbb{C}$ as a group of translations;
4. $\mathbb{C}/\Lambda$, where $\Lambda = \mathbb{Z} + \tau\mathbb{Z}$ is a lattice with $\tau \in \mathbb{H}^2$ acting on $\mathbb{C}$ as a group of translations;
5. $\mathbb{H}^2/\Gamma$, where Γ is a subgroup of $\mathrm{Aut}(\mathbb{H}^2)$ that acts properly discontinuously and freely.

$\boxed{84}$ **Corollary.** Any two compact Riemann surfaces of genus 0 are biholomorphic.

§ The Teichmüller Space of Genus 1.

$\boxed{85}$ **Theorem.** Let $\tau, \tau' \in \mathbb{H}$, and consider the following Riemann surfaces of genus 1:

$$\mathbb{T}_\tau := \mathbb{C}/\langle z \mapsto z+1,\ z \mapsto z+\tau\rangle \qquad \text{and} \qquad \mathbb{T}_{\tau'} := \mathbb{C}/\langle z \mapsto z+1,\ z \mapsto z+\tau'\rangle.$$

Then $\mathbb{T}_\tau$ and $\mathbb{T}_{\tau'}$ are biholomorphic if and only if

$$\tau' = \frac{a\tau + b}{c\tau + d}. \tag{$*$}$$

for some $a, b, c, d \in \mathbb{Z}$ satisfying $ad - bc = 1$.

Proof. Suppose $f\colon \mathbb{T}_{\tau'} \to \mathbb{T}_\tau$ is a biholomorphism. Then there exists an automorphism $\widetilde{f}\colon \mathbb{C} \to \mathbb{C}$ given by $\widetilde{f}(z) = \alpha z + \beta$ for which the diagram commutes

$$\begin{tikzcd}
\mathbb{C} \arrow[r, "\widetilde{f}"] \arrow[d, "z\mapsto[z]"'] & \mathbb{C} \arrow[d, "z\mapsto[z]"] \\
\mathbb{T}_{\tau'} \arrow[r, "f"] & \mathbb{T}_\tau
\end{tikzcd}$$

Because f is a well-defined bijection, $\widetilde{f}(\mathbb{Z} + \tau'\mathbb{Z}) = \mathbb{Z} + \tau\mathbb{Z} + \beta$.

Hence, $\widetilde{f}(\tau') - \beta = \alpha\tau'$ and $\widetilde{f}(1) - \beta = \alpha$ must be generators of $\mathbb{Z} + \tau\mathbb{Z}$. Thus, $\alpha\tau' = a\tau + b$ and $\alpha = c\tau + d$ for some $a, b, c, d \in \mathbb{Z}$ satisfying $ad - bc = 1$. Dividing these two equations yields $(*)$.

Conversely, if $(*)$ holds, then a biholomorphic mapping $\mathbb{T}_{\tau'} \to \mathbb{T}_\tau$ is given by

$$[z] \longmapsto [(c\tau + d)z]. \qquad \square$$

$\boxed{86}$ **Corollary.** We can parametrize all isomorphism classes of complex structures on the torus by the set

$$\mathscr{M}_1 := \mathbb{H}^2 / \operatorname{PSL}(2, \mathbb{Z}).$$

$\boxed{87}$ **Note.** A *hyperbolic 2-orbifold* Q is a quotient $\mathbb{H}^2/\Gamma$, where Γ is a *Fuchsian group* (a discrete subgroup of the Lie group $\operatorname{Aut}(\mathbb{H}^2) \equiv \operatorname{PSL}(2, \mathbb{R})$).

- For example, $\mathscr{M}_1$ is a hyperbolic 2-orbifold.
- It can be shown that if Γ is torsion-free (i.e., has no elements of finite order), then Q is a manifold.

88 **Definition.** Let Γ be a Fuchsian group. A *fundamental domain* for Γ is a connected subset $\mathcal{F} \subset \mathbb{H}^2$ such that $\mathcal{F} = \overline{\mathrm{int}(\mathcal{F})}$ and:

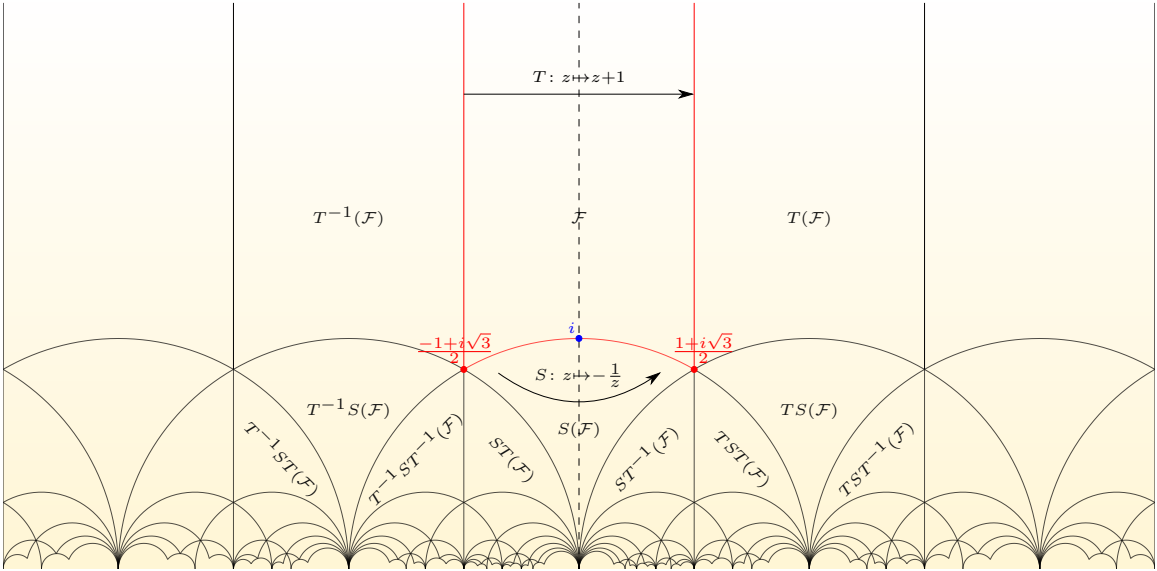
- $\bigcup_{\gamma \in \Gamma} \gamma(\mathcal{F}) = \mathbb{H}^2$,
- $\mathrm{int}(\mathcal{F}) \cap \gamma(\mathrm{int}(\mathcal{F})) = \varnothing$ for all $\gamma \in \Gamma \setminus \{\mathrm{Id}\}$.

89 **Theorem** (*Dirichlet Domain* [Kat1992]). Let Γ be a Fuchsian group, and let $p \in \mathbb{H}^2$ be a point that is not fixed by any element of Γ (except the identity). For each $\gamma \in \Gamma \setminus \{\mathrm{Id}\}$, let $H_p(\gamma)$ be the closed hyperbolic half-plane containing p bounded by the perpendicular bisector of the geodesic segment $[p, \gamma(p)]$. Then the set

$$D_p(\Gamma) := \bigcap_{\gamma \in \Gamma \setminus \{\mathrm{Id}\}} H_p(\gamma)$$

is a fundamental domain for Γ.

90 **Example.** [Kat1992] A fundamental domain for $\mathrm{PSL}(2, \mathbb{Z})$ is given by $D_{2i}(\mathrm{PSL}(2, \mathbb{Z})) = \{z \in \mathbb{H}^2 : |z| \geq 1,\ |\mathrm{Re}(z)| \leq \frac{1}{2}\}$.



The family $\{\gamma(\mathcal{F}) \mid \gamma \in \mathrm{PSL}(2,\mathbb{Z})\}$ is a tessellation of $\mathbb{H}^2$

$\boxed{91}$ **Corollary.** The hyperbolic 2-orbifold $\mathscr{M}_1$ is homeomorphic to $\mathbb{C}$.

Proof. The automorphism $T\colon z \mapsto z+1$ maps the line $\mathrm{Re}(z) = -1/2$ onto the line $\mathrm{Re}(z) = 1/2$. Furthermore, the automorphism $S\colon z \mapsto -1/z$ maps the unit circular arc joining $-1/2 + i\sqrt{3}/2$ to i onto the unit circular arc joining i to $1/2 + i\sqrt{3}/2$. $\square$

$\boxed{92}$ **Theorem.** [Nog[1998]; Gir+[2012]] Let $\tau \in \mathbb{H}^2$, and define $\Lambda := \mathbb{Z} + \tau\mathbb{Z}$. Then the series

$$\wp(z) = \frac{1}{z^2} + \sum_{0 \neq \omega \in \Lambda} \left(\frac{1}{(z-\omega)^2} - \frac{1}{\omega^2} \right)$$

defines a meromorphic function in $\mathbb{C}$, called the *Weierstrass $\wp$ function*, such that $\wp(z+\omega) = \wp(z)$ and $\wp'(z+\omega) = \wp'(z)$ for all $z \in \mathbb{C}$ and all $\omega \in \Lambda$. Moreover, we have the following:

- If we set $g_2(\tau) := 60 \sum_{0 \neq \omega \in \Lambda} \omega^{-4}$ and $g_3(\tau) := 140 \sum_{0 \neq \omega \in \Lambda} \omega^{-6}$, then for all $z \in \mathbb{C}$,

$$\wp'(z)^2 = 4\wp(z)^3 - g_2(\tau)\wp(z) - g_3(\tau).$$

- The Riemann surface of the polynomial $F(z,w) = w^2 - 4z^3 + g_2(\tau)z + g_3(\tau)$ is biholomorphic to $\mathbb{C}/\Lambda$.
- The function $g_2\colon \mathbb{H}^2 \to \mathbb{C}$ is holomorphic and satisfies $g_2\left(\frac{a\tau+b}{c\tau+d}\right) = (c\tau+d)^4 g_2(\tau)$ for every $\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \mathrm{SL}(2,\mathbb{Z})$, i.e., g_2 is a *modular form* of weight 4. Analogously, g_3 is a modular form of weight 6.
- The *absolute elliptic invariant* $J\colon \mathbb{H}^2 \to \mathbb{C}$ defined by

$$J(\tau) := \frac{g_2(\tau)^3}{g_2(\tau)^3 - 27g_3(\tau)^2}$$

is an $\mathrm{SL}(2,\mathbb{Z})$-invariant, surjective, holomorphic map. Moreover, the map

$$\mathscr{M}_1 := \mathbb{H}^2/\,\mathrm{PSL}(2,\mathbb{Z}) \to \mathbb{C} \quad \text{defined by} \quad [\tau] \mapsto J(\tau)$$

is a homeomorphism.

§ Teichmüller Space of Genus g.

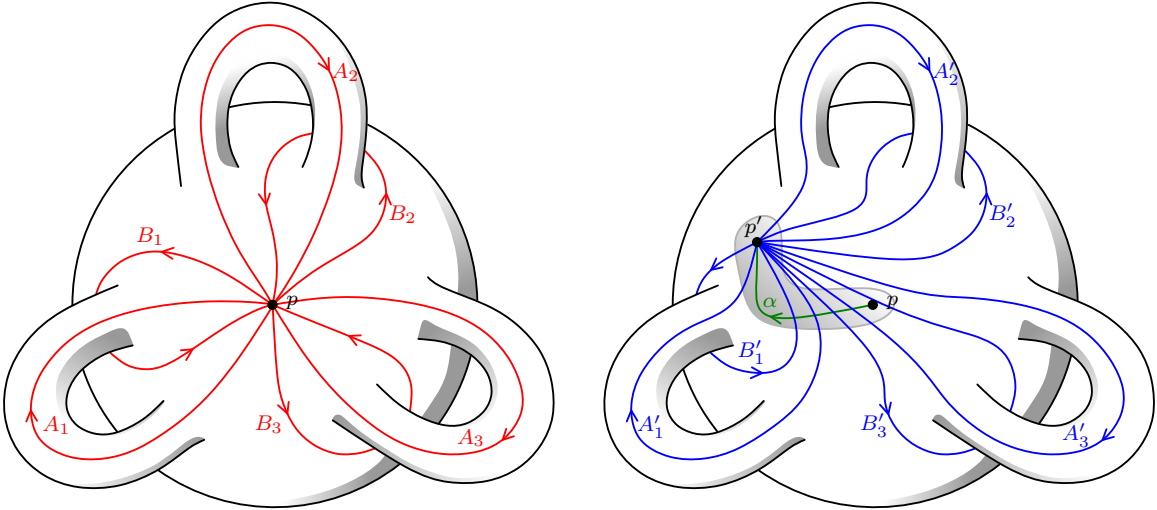
Let R be a compact Riemann surface of genus $g \geq 1$, and let p be a point in R.

$\boxed{93}$ **Definition.** An ordered $2g$-tuple $\mathscr{B}_p := (a_1, b_1, \ldots, a_g, b_g)$ of homotopy classes in $\pi_1(R, p)$ is called a *marking on R* if it generates $\pi_1(R, p)$ and satisfies the single commutator relation:

$$\prod_{i=1}^{g} a_i b_i a_i^{-1} b_i^{-1} = 1$$

$\boxed{94}$ **Theorem.** [Nie1927] [Zie+1980] Let $\alpha\colon \pi_1(R, p) \to \pi_1(R, p)$ be an automorphism. Then there exists a homeomorphism $\varphi\colon R \to R$ with $\varphi(p) = p$ such that $\varphi_* = \alpha$.

$\boxed{95}$ **Corollary.** A marking $\mathscr{B}_p$ on R can be represented by a collection of simple closed curves $A_i \in a_i$ and $B_i \in b_i$ based at p that intersect each other only at the basepoint p, and whose complement $R \setminus \bigcup_{i=1}^{g} (A_i \cup B_i)$ is homeomorphic to an open disk.



The red loops (left) and the blue loops (right) yield two equivalent markings

$\boxed{96}$ **Definition.** Two markings $\mathscr{B}_p := (a_1, b_1, \ldots, a_g, b_g)$ and $\mathscr{B}_{p'} := (a'_1, b'_1, \ldots, a'_g, b'_g)$ on R are called *equivalent* if there exists a path $\alpha\colon [0,1] \to R$ from p to p' such that the base change isomorphism $\pi_1(R,p) \to \pi_1(R,p')$ defined by $[\ell] \mapsto [\overline{\alpha} * \ell * \alpha]$ sends a_i to a'_i and b_i to b'_i for all $i = 1, \ldots, g$.

$\boxed{97}$ **Definition.** [Ima+1992] Let $\mathscr{X}_g$ be the collection of all pairs $(R, \mathscr{B}_p)$, where R is a compact Riemann surface of genus g and $\mathscr{B}_p = (a_1, b_1, \ldots, a_g, b_g)$ is a marking on R based at p.

Two pairs $(R, \mathscr{B}_p)$ and $(S, \mathscr{B}_q) \in \mathscr{X}_g$, with $\mathscr{B}_q = (c_1, d_1, \ldots, c_g, d_g)$ based at $q \in S$, are said to be *equivalent* if there exists a biholomorphism $h\colon S \to R$ such that the marking

$$h_*(\mathscr{B}_q) := (h_*(c_1), h_*(d_1), \ldots, h_*(c_g), h_*(d_g))$$

is equivalent to $\mathscr{B}_p$ in the sense of equivalent markings on R.

The *Teichmüller space $\mathscr{T}_g$ of genus g* is the set of equivalence classes of $\mathscr{X}_g$ under this relation. An equivalence class $[R, \mathscr{B}_p] \in \mathscr{T}_g$ is called a *marked compact Riemann surface of genus g*.

§ Teichmüller Space of Genus 1 and the Upper Half-Plane.

Setup. Let $\tau \in \mathbb{H}^2$, and consider the segments $t \mapsto t$ and $t \mapsto t\tau$ for $t \in [0,1]$ in $\mathbb{C}$.

Under the projection map $\mathbb{C} \to \mathbb{T}_\tau$, these segments give two simple closed curves $A_1(\tau)$ and $B_1(\tau)$ in $\mathbb{T}_\tau$ based at $[0]$, respectively.

Define $\mathscr{B}_{[0]}(\tau)$ as the ordered pair $([A_1(\tau)], [B_1(\tau)])$ of the homotopy classes of the simple closed curves $A_1(\tau)$ and $B_1(\tau)$. Thus, $\mathscr{B}_{[0]}(\tau)$ is a marking on $\mathbb{T}_\tau$.

$\boxed{98}$ **Theorem.** [Ima+1992] Every element of the Teichmüller space T_1 of genus 1 can be represented by a pair of the form $(\mathbb{T}_\tau, \mathscr{B}_{[0]}(\tau))$ for some $\tau \in \mathbb{H}^2$.

Moreover, two pairs $(\mathbb{T}_\tau, \mathscr{B}_{[0]}(\tau))$ and $(\mathbb{T}_{\tau'}, \mathscr{B}_{[0]}(\tau'))$ give the same element in T_1 if and only if $\tau = \tau'$.

$\boxed{99}$ **Corollary.** There is a bijective correspondence between $\mathbb{H}^2$ and $\mathscr{T}_1$.

§ Teichmüller Spaces of Riemann Surfaces.

100 Definition. Fix a compact Riemann surface R. The *Teichmüller space* of R, denoted $\mathrm{Teich}(R)$, is defined as

$$\mathrm{Teich}(R) := \left\{ (S, f) \;\middle|\; \begin{array}{l} S \text{ is a compact Riemann surface, and} \\ f\colon R \to S \text{ is an orientation-preserving diffeomorphism} \end{array} \right\} \Big/ \sim$$

where two pairs (S, f) and (S', g) are equivalent if the map $g \circ f^{-1}\colon S \to S'$ is homotopic to a biholomorphism $h\colon S \to S'$.

101 Theorem. [Ima+1992] Let R be a compact Riemann surface of genus g with a marking $\mathscr{B}_p = (a_1, b_1, \ldots, a_g, b_g)$ based at the point p. Then the map $\Phi_{\mathscr{B}_p}\colon \mathrm{Teich}(R) \to \mathscr{T}_g$ defined by

$$[S, f] \longmapsto [S, (f_*(a_1), f_*(b_1), \ldots, f_*(a_g), f_*(b_g))]$$

is a well-defined bijection.

$\boxed{102}$ **Theorem.** Let R be a compact Riemann surface, and let $\mathrm{Mod}(R)$ be the mapping class group of R, identified with the group of homotopy classes of orientation-preserving diffeomorphisms $\omega\colon R \to R$. Then the action of $\mathrm{Mod}(R)$ on $\mathrm{Teich}(R)$ defined by

$$[\omega] \cdot [S, f] := \left[S, f \circ \omega^{-1}\right]$$

is well-defined.

$\boxed{103}$ **Theorem.** Let R be a compact Riemann surface of genus g, and let $\mathscr{M}_g$ be the set of all biholomorphic equivalence classes $[S]$ of compact Riemann surfaces S of genus g, called the *moduli space of compact Riemann surfaces of genus g*. Then the natural map $\mathrm{Teich}(R) \to \mathscr{M}_g$ given by $[S, f] \longmapsto [S]$ descends to a canonical bijection:

$$\mathrm{Teich}(R)/\,\mathrm{Mod}(R) \xrightarrow{\cong} \mathscr{M}_g.$$

§ Nielsen Realization Problem.

Let R be a compact Riemann surface of genus $g \geq 2$.

104 Theorem. [Mar[2007]; Mor[1987]; Mor[2001]] The exact sequence

$$1 \longrightarrow \mathrm{Diff}_0(R) \longrightarrow \mathrm{Diff}^+(R) \xrightarrow{\pi} \underbrace{\mathrm{Diff}^+(R)/\mathrm{Diff}_0(R)}_{=\mathrm{Mod}(R)} \longrightarrow 1 \qquad (*)$$

does not split in general.

105 Question. [Nie[1943]] For which subgroups $G \subset \mathrm{Mod}(R)$ does there exist a homomorphism $s_G \colon G \to \mathrm{Diff}^+(R)$ such that $\pi \circ s_G = \mathrm{Id}_G$?

106 Theorem. [Sch[1879]] The group $\mathrm{Aut}(R)$ is finite.

107 Theorem. [Hur[1892]] The order of $\mathrm{Aut}(R)$ is at most $84(g-1)$.

108 **Theorem.** [Das+2007] Let ψ_1 and ψ_2 be two homotopic automorphisms of R. Then $\psi_1 = \psi_2$.

109 **Corollary** (Fenchel's observation)**.** Let $[S, f] \in \mathrm{Teich}(R)$, and let $G \leq \mathrm{Mod}(R)$ be a subgroup such that $[\omega] \cdot [S, f] = [S, f]$ for all $[\omega] \in G$. Then we have the following:

- For each diffeomorphism $\omega \in \mathrm{Diff}^+(R)$ representing a class in G, the map $f \circ \omega \circ f^{-1}$ is homotopic to a unique element $h_\omega \in \mathrm{Aut}(S)$.
- The assignment $\Phi \colon G \to \mathrm{Aut}(S)$ given by $[\omega] \longmapsto h_\omega$ is a well-defined, injective group homomorphism.
- The map $s_G \colon G \to \mathrm{Diff}^+(R)$ given by $s_G([\omega]) = f^{-1} \circ \Phi([\omega]) \circ f$ is a well-defined, injective group homomorphism. Moreover, $s_G([\omega])$ represents the class $[\omega]$ for every $[\omega] \in G$.

110 **Theorem.** [Ker1983] [Wol1987] Let G be a finite subgroup of $\mathrm{Mod}(R)$. Then G has a fixed point in $\mathrm{Teich}(R)$. Thus, the sequence $(*)$ splits over all finite subgroups of $\mathrm{Mod}(R)$.

§ The Dimension of the Moduli Space for $g > 1$.

$\boxed{111}$ **Theorem.** [Gir+2012] Let Γ_1 and Γ_2 be two Fuchsian groups whose actions on $\mathbb{H}^2$ are free. Then $\mathbb{H}^2/\Gamma_1$ and $\mathbb{H}^2/\Gamma_2$ are biholomorphic if and only if there exists $T \in \mathrm{PSL}(2, \mathbb{R})$ such that $T\Gamma_1 T^{-1} = \Gamma_2$.

$\boxed{112}$ **Counting.** [Gir+2012] Let R be a compact Riemann surface of genus $g \geq 2$. By the Uniformization Theorem, $R \cong \mathbb{H}^2/\Gamma$ for a Fuchsian group $\Gamma \cong \pi_1(R, *)$.

Thus, describing R amounts to specifying a set $\mathscr{S} = \{\gamma_1, \delta_1, \ldots, \gamma_g, \delta_g\}$ of $2g$ real 2×2 matrices such that:

(i) $\det(\gamma_i) = 1 = \det(\delta_i)$ for all $i = 1, \ldots, g$.

(ii) $\prod_{i=1}^{g} \gamma_i \delta_i \gamma_i^{-1} \delta_i^{-1} = \left(\begin{smallmatrix} 1 & 0 \\ 0 & 1 \end{smallmatrix}\right)$.

By (i), the $2g$ matrices yield $6g$ real parameters.

Since the product $\prod_{i=1}^{g} \gamma_i \delta_i \gamma_i^{-1} \delta_i^{-1}$ in (ii) has determinant 1, setting it to the identity imposes exactly 3 constraints, leaving $6g - 3$ real parameters.

Factoring out conjugation by $\mathrm{PSL}(2, \mathbb{R})$ subtracts $\dim \mathrm{PSL}(2, \mathbb{R}) = 3$. Therefore, $\mathscr{M}_g$ requires $6g - 6$ real parameters.

History

[Rie1857] Bernhard Riemann: Introduced "moduli" and determined that $\mathscr{M}_g$ depends on $3g - 3$ complex parameters.

[Tei1940] Oswald Teichmüller: Introduced topological markings to define the simply-connected space $\mathrm{Teich}(R)$.

[Fen+1940] Werner Fenchel and Jakob Nielsen: Constructed global coordinates providing a diffeomorphism from $\mathrm{Teich}(R)$ to $\mathbb{R}^{6g-6}$.

[Ahl1954] Lars Ahlfors: Rigorously established the complex analytic and smooth structures of $\mathrm{Teich}(R)$.

[Ber1960] Lipman Bers: Proved $\mathrm{Teich}(R)$ embeds as a bounded, holomorphically contractible domain in $\mathbb{C}^{3g-3}$.

[Del+1969] Pierre Deligne and David Mumford: Formalized the global algebraic structure of moduli space and its compactification $\overline{\mathscr{M}_g}$.

Metrics, Geodesics, and Curvature: Quick Overview via Hyperbolic Plane

§ Riemannian Metrics: Definition and Examples.

$\boxed{113}$ **Definition.** Let M be a smooth manifold. A *Riemannian metric* on M is an assignment $p \longmapsto g_p$ that smoothly associates to each point $p \in M$ a positive-definite inner product g_p on T_pM. Smoothness means that in any chart $(U, (x^1, \ldots, x^n))$, the functions

$$g_{ij}(p) := g_p\left(\left.\frac{\partial}{\partial x^i}\right|_p, \left.\frac{\partial}{\partial x^j}\right|_p\right)$$

are smooth functions on U for all indices i, j.

$\boxed{114}$ **Example.** Let $M = \mathbb{R}^n$ with standard coordinates $(x^1, \ldots, x^n)$. The standard Euclidean metric g is defined by $g_{ij}(p) := \delta_{ij}$ for all $p \in \mathbb{R}^n$. The metric can be expressed globally as

$$g = dx^1 \otimes dx^1 + \cdots + dx^n \otimes dx^n.$$

115 Theorem. [Spi[1979][1]] Every smooth manifolds admits a (non-canonical) Riemannian metric.

116 Example. (Beltrami) Let $\mathbb{H}^2 := \{(x, y) \in \mathbb{R}^2 \mid y > 0\}$ be the upper half-plane with standard global coordinates (x, y). The hyperbolic metric g is defined by $g_{ij}(p) = \frac{1}{y^2}\delta_{ij}$ for all $p = (x, y) \in \mathbb{H}^2$. The metric can be expressed globally as

$$g = \frac{1}{y^2}(dx \otimes dx + dy \otimes dy).$$

117 Example. Let $n \geq 2$, and let $\mathbb{H}^n := \{(x^1, \ldots, x^n) \in \mathbb{R}^n \mid x^n > 0\}$ be the upper half-space with standard global coordinates $(x^1, \ldots, x^n)$. The hyperbolic metric g is defined by $g_{ij}(p) = \frac{1}{(x^n)^2}\delta_{ij}$ for all $p = (x^1, \ldots, x^n) \in \mathbb{H}^n$. The metric can be expressed globally as

$$g = \frac{1}{(x^n)^2}\left(dx^1 \otimes dx^1 + \cdots + dx^n \otimes dx^n\right).$$

§ Riemannian Distance: The Induced Distance Function.

118 Definition. Let (M, g) be a Riemannian manifold. Define the *length* of a smooth curve $c\colon [a, b] \to M$ as:

$$L(c) := \int_a^b \|c'(t)\|\, dt \qquad \text{where} \qquad \|c'(t)\| := \sqrt{g_{c(t)}\left(c'(t), c'(t)\right)}.$$

The length of a continuous and piecewise smooth curve may be defined as the sum of the lengths of its smooth segments.

119 Theorem. [Spi1979[1]] Let (M, g) be a connected Riemannian manifold. The function $d\colon M \times M \to [0, \infty)$ defined by

$$d(p, q) := \inf \left\{ L(c) \mid c\colon [a, b] \to M \text{ is a piecewise smooth curve with } c(a) = p, c(b) = q \right\},$$

is a metric on M, called the *Riemannian distance*, and the topology induced by (M, d) (as a metric space) coincides with the manifold topology of M.

122 **Theorem.** (Cayley-Klein) [Kat1992] The Riemannian (hyperbolic) distance between points z and w in $\mathbb{H}^2(\subset \mathbb{C})$ is given by:

$$d(z,w) = \ln \frac{|z-\overline{w}| + |z-w|}{|z-\overline{w}| - |z-w|}.$$

123 **Corollary.** As $t \to 0^+$, $d(i, it)$ diverges to ∞, whereas $|i - it| \to 1$.

§ Christoffel Symbols of the First and Second Kinds.

124 **Definition.** [Spi1979 1; Spi1979 2] Let $(U, \varphi = (x^1, \ldots, x^n))$ be a chart of the Riemannian manifold (M, g). The *Christoffel symbols* with respect to φ are the smooth functions $[ij, l] \colon U \to \mathbb{R}$ and $\Gamma^k_{ij} \colon U \to \mathbb{R}$ defined for all $i, j, k, l \in \{1, \ldots, n\}$ by the equations:

$$[ij, l] := \frac{1}{2}\left(\frac{\partial g_{il}}{\partial x^j} + \frac{\partial g_{jl}}{\partial x^i} - \frac{\partial g_{ij}}{\partial x^l}\right) \qquad \text{and} \qquad \Gamma^k_{ij} := \sum_{l=1}^{n} g^{kl} \cdot [ij, l],$$

where g^{kl} denotes the components of the inverse metric matrix (i.e., $\sum_{l=1}^{n} g^{kl} g_{lj} = \delta^k_j$).

125 **Note.** [Jos[2008]; Spi[1979][1]] Christoffel symbols arise naturally both in the existence and uniqueness theorem for the *Levi–Civita connection* and in the calculus of variations.

§ Geodesics: Locally Distance-Minimizing Constant-Speed Curves.

126 **Definition.** [Jos[2008]; Spi[1979][1]] Let $\gamma\colon I \to M$ be a smooth curve in a Riemannian manifold (M, g) defined on an interval I. Then γ is said to be a *geodesic* if for every chart $(U, (x^1, \ldots, x^n))$, the coordinate functions $\gamma^k := x^k \circ \gamma|_{\gamma^{-1}(U)}$ satisfy the *Euler–Lagrange geodesic equations*

$$\frac{d^2\gamma^k}{dt^2}(t) + \sum_{i,j=1}^{n} \Gamma^k_{ij}(\gamma(t))\frac{d\gamma^i}{dt}(t)\frac{d\gamma^j}{dt}(t) = 0,$$

for all $t \in \gamma^{-1}(U)$ and for each $k = 1, \ldots, n$.

127 **Proposition.** (Picard-Lindelöf) [Lee[2018]] Given $p \in M$ and $v \in T_pM$, there exists a unique geodesic $\gamma_{p,v}\colon I \to M$, defined on a maximal open interval I containing 0, such that $\gamma_{p,v}(0) = p$ and $\gamma'_{p,v}(0) = v$.

$\boxed{128}$ **Proposition.** [Spi[1979[1]]] Let γ be a geodesic in a Riemannian manifold (M, g) defined on an interval I. Then there exists a constant k such that $\|\gamma'(t)\| = k$ for all $t \in I$.

$\boxed{129}$ **Proposition.** [Lee[2018]] Let (M, g) be a connected Riemannian manifold, and let d be its Riemannian distance. Then for each $p \in M$, there exists a neighborhood U_p such that for every $q \in U_p$, there is a unique geodesic $\gamma\colon [0, \ell] \to M$ from p to q such that

- $\|\gamma'(t)\| = 1$ for all $t \in [0, \ell]$ and
- $d(p, q) = \ell = L(\gamma)$.

§ Geodesics in the Upper Half-Plane Model.

Consider the hyperbolic plane $\mathbb{H}^2$. For $p = (x, y) \in \mathbb{H}^2$, we have

$$(g_{ij}(p)) = \begin{pmatrix} 1/y^2 & 0 \\ 0 & 1/y^2 \end{pmatrix} \qquad \text{and} \qquad \left(g^{ij}(p)\right) = \begin{pmatrix} y^2 & 0 \\ 0 & y^2 \end{pmatrix}.$$

Thus, the only nonzero Christoffel symbols of the second kind are

$$\Gamma^1_{12}(p) = \Gamma^1_{21}(p) = -1/y, \quad \Gamma^2_{11}(p) = 1/y, \quad \text{and} \quad \Gamma^2_{22}(p) = -1/y.$$

Let $\gamma(t) = (x(t), y(t))$ be a non-constant geodesic. Then the system of geodesic equations consists of

$$\ddot{x} - \frac{2}{y}\dot{x}\dot{y} = 0 \tag{$*$}$$

$$\ddot{y} + \frac{1}{y}\dot{x}^2 - \frac{1}{y}\dot{y}^2 = 0. \tag{$**$}$$

By a simple reparametrization, we may assume that $\|\gamma'(t)\| = 1$ for all t, i.e.,

$$\dot{x}^2 + \dot{y}^2 = y^2. \tag{$***$}$$

- If $\dot{x} \equiv 0$, then x is constant.
- If $\dot{x} \not\equiv 0$, then $(*)$ is equivalent to $\ddot{x}/\dot{x} = 2\dot{y}/y$, which gives $\dot{x} = ky^2$ for some $k \in \mathbb{R}$. Plugging this into $(***)$, and solving, we get $(x-d)^2 + y^2 = 1/k^2$ for some $d \in \mathbb{R}$.
- Equation $(**)$ can be derived from equations $(*)$ and $(***)$ [Tu2017].

130 **Theorem.** We have the following observations on the geodesics of $\mathbb{H}^2$:

- *Geodesic Connectedness:* Given any two points in $\mathbb{H}^2$, there exists a geodesic passing through them.
- *Geodesic Completeness:* Every geodesic can be extended to a geodesic defined on all of $\mathbb{R}$.
- *The Hyperbolic Parallel Postulate:* Given a point $p \in \mathbb{H}^2$ and a geodesic γ not passing through p, there are infinitely many geodesics passing through p that do not intersect γ.

§ The Hopf–Rinow Theorem: Connecting Two Points with a Distance-Minimizing Geodesic.

131 **Theorem.** [Spi[1979][1]] Let (M, g) be a connected Riemannian manifold, and let d be its Riemannian distance. The following are equivalent:

(1) Every geodesic can be extended to a geodesic defined on all of $\mathbb{R}$.

(2) The metric space (M, d) is complete.

(3) A subset $A \subset M$ is compact if and only if it is closed and bounded with respect to d.

If (1) (hence all) of the above conditions hold, then M is said to be *geodesically complete*. In this case, for every $p, q \in M$, there exists a geodesic $\gamma\colon [0, \ell] \to M$ from p to q such that

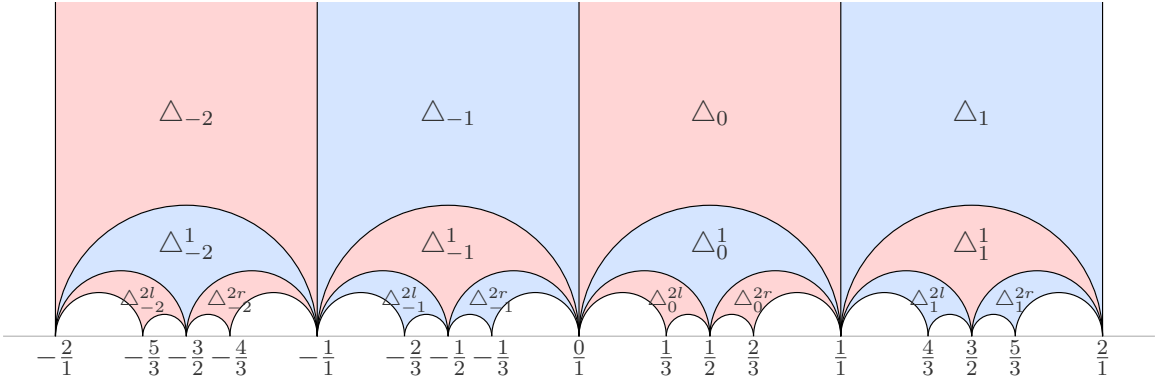
- $\|\gamma'(t)\| = 1$ for all $t \in [0, \ell]$,
- $d(p, q) = \ell = L(\gamma)$.

§ The Farey Tessellation: A Tiling of the Upper Half-Plane by Ideal Triangles.

$\boxed{132}$ **Theorem.** [Hat2022] The hyperbolic plane $\mathbb{H}^2$ can be written as the union of a countable collection $\mathfrak{F}$ of ideal geodesic triangles with pairwise disjoint interiors such that the following hold for every $\triangle \in \mathfrak{F}$:

- The three vertices of $\triangle$ are elements of $\widehat{\mathbb{Q}} := \mathbb{Q} \cup \{\frac{1}{0}\}$.

- There exist two vertices of $\triangle$ with lowest-term expressions $\frac{p}{q}$ and $\frac{r}{s}$ satisfying $|ps - qr| = 1$, whose *Farey sum* $\frac{p}{q} \oplus \frac{r}{s} := \frac{p+r}{q+s}$ exactly yields the third vertex of $\triangle$.



Inductive construction of the Farey tessellation, where every $\triangle \in \mathfrak{F}$ is equal to $T(\triangle_0)$ for some $T \in \mathrm{PSL}(2, \mathbb{Z})$.

§ Local and Global Isometries.

133 Definition. Let (M, g) and (N, h) be Riemannian manifolds. A diffeomorphism $f\colon M \to N$ is called an *isometry* if $f^*h = g$, that is, for any point $p \in M$ and tangent vectors $u, v \in T_pM$, we have

$$h_{f(p)}(df_p(u), df_p(v)) = g_p(u, v).$$

A map $\varphi\colon M \to N$ is called a *local isometry* if for each point $p \in M$ there exists a neighborhood U of p such that $\varphi|_U$ is an isometry onto an open subset of N.

134 Theorem. [Ser2013] Let $\mathrm{Isom}^+(\mathbb{H}^2)$ be the group of orientation-preserving isometries of $\mathbb{H}^2$. Then every element of $\mathrm{Isom}^+(\mathbb{H}^2)$ is of the form

$$z \longmapsto (az + b)/(cz + d), \quad \text{where } a, b, c, d \in \mathbb{R} \text{ and } ad - bc = 1.$$

Every orientation-reversing isometry of $\mathbb{H}^2$ is of the form $f \circ r$, where $f \in \mathrm{Isom}^+(\mathbb{H}^2)$ is an orientation-preserving isometry, and r is the isometry of $\mathbb{H}^2$ defined by $r(z) := -\overline{z}$.

$\boxed{135}$ **Theorem.** [Kat1992] The map from $\mathrm{Isom}^+(\mathbb{H}^2)$ to the unit tangent bundle $U\mathbb{H}^2$ given by

$$T \longmapsto (T(i), dT_i(0,1))$$

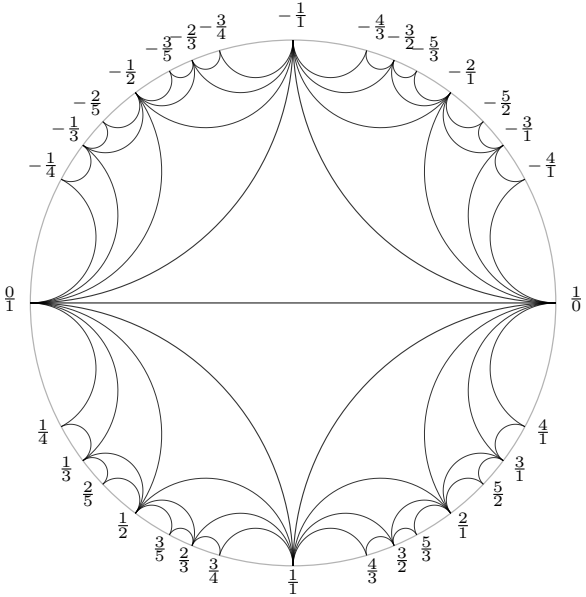
is a diffeomorphism. Thus, given two unit-speed geodesics $\gamma, \delta\colon [0,\ell] \to \mathbb{H}^2$, there exists a unique $T \in \mathrm{Isom}^+(\mathbb{H}^2)$ such that $T \circ \gamma = \delta$.

§ Poincaré Ball Model.

$\boxed{136}$ **Definition.** Let $n \geq 2$, and let $\mathbb{B}^n := \{(x^1, \ldots, x^n) \in \mathbb{R}^n \mid (x^1)^2 + \cdots + (x^n)^2 < 1\}$. The hyperbolic metric g on $\mathbb{B}^n$ is defined globally by

$$g = \frac{4}{(1-(x^1)^2 - \cdots - (x^n)^2)^2}\left(dx^1 \otimes dx^1 + \cdots + dx^n \otimes dx^n\right).$$

$\boxed{137}$ **Theorem.** [Ser2013] The Cayley transformation $\mathbb{H}^2 \to \mathbb{B}^2$ given by $z \longmapsto (z-i)/(z+i)$ is an isometry.



Inductive construction of the Farey tessellation in the Poincaré disk model

§ Hyperbolic Structures.

$\boxed{138}$ **Definition.** Let S be a connected, orientable, topological 2-manifold without boundary. An $(\mathbb{H}^2, \mathrm{Isom}^+(\mathbb{H}^2))$-*atlas* on S is a family of functions $\{\varphi_i \colon U_i \to \mathbb{H}^2\}_{i \in \mathcal{I}}$, called *charts*, satisfying the following conditions:

- The set U_i is an open connected subset of S for each $i \in \mathcal{I}$.
- The chart φ_i maps U_i homeomorphically onto an open subset of $\mathbb{H}^2$ for each $i \in \mathcal{I}$.
- The collection $\{U_i\}_{i \in \mathcal{I}}$ is a cover S.
- If $U_i \cap U_j \neq \varnothing$, then the transition function

$$\Phi_{ij} \coloneqq \varphi_j \circ \varphi_i^{-1} \colon \varphi_i(U_i \cap U_j) \to \varphi_j(U_i \cap U_j)$$

  agrees in a neighborhood of each point of its domain with an element of $\mathrm{Isom}^+(\mathbb{H}^2)$.

A *hyperbolic structure* on S is a *maximal* $(\mathbb{H}^2, \mathrm{Isom}^+(\mathbb{H}^2))$-*atlas*.

$\boxed{139}$ **Theorem.** Suppose S admits a hyperbolic structure. Let $g_{\mathbb{H}^2}$ be the hyperbolic metric on $\mathbb{H}^2$. For every chart $\varphi_i \colon U_i \to \mathbb{H}^2$, define a local metric g_i on U_i via pullback:

$$g_i \coloneqq \varphi_i^* g_{\mathbb{H}^2}.$$

If $U_i \cap U_j \neq \varnothing$, then $(g_i)_p = (g_j)_p$ for all $p \in U_i \cap U_j$. Thus, the collection $\{g_i\}$ defines a globally well-defined Riemannian metric on S, called the *hyperbolic metric associated to the hyperbolic structure on S*.

$\boxed{140}$ **Example.** Any connected open subset $U \subset \mathbb{H}^2$ admits a hyperbolic structure. Consider the atlas consisting of exactly one chart, namely the inclusion map.

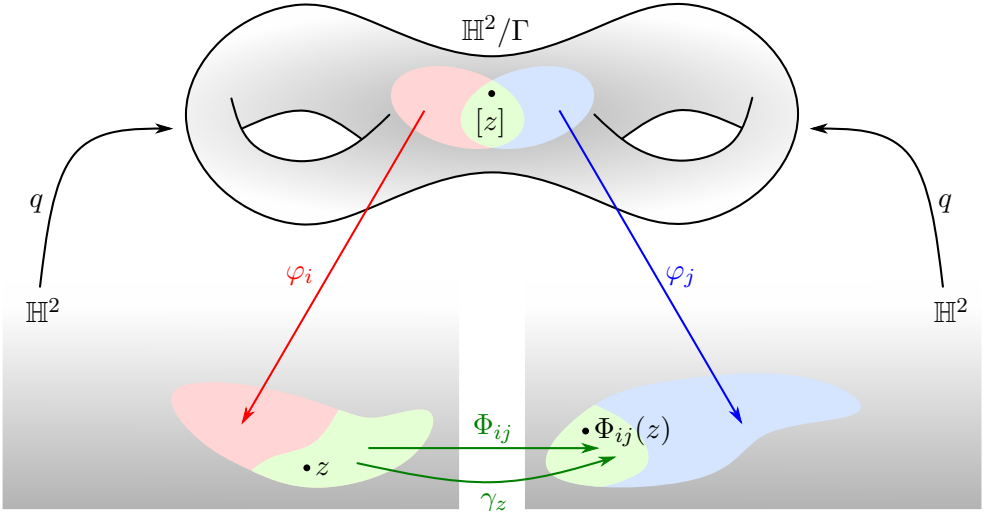
$\boxed{141}$ **Remark.** The associated hyperbolic metric on a connected open subset U of $\mathbb{H}^2$ may not be geodesically complete in general. For instance, $\{(x, y) \in \mathbb{H}^2 \mid x^2 + (y-1)^2 < 1/4\}$ is not geodesically complete.

$\boxed{142}$ **Definition.** A subgroup Γ of $\mathrm{Isom}^+(\mathbb{H}^2) \equiv \mathrm{PSL}(2,\mathbb{R})$ is called a *Fuchsian group* if there exists an open neighborhood U of the identity I in $\mathrm{PSL}(2,\mathbb{R})$ such that $U \cap \Gamma = \{\mathrm{I}\}$.

$\boxed{143}$ **Theorem.** [Kat1992] [Ser2013] A subgroup G of $\mathrm{PSL}(2,\mathbb{R})$ is a Fuchsian group if and only if its action on $\mathbb{H}^2$ is properly discontinuous.

$\boxed{144}$ **Theorem.** Let Γ be a Fuchsian group. Suppose the action of Γ on $\mathbb{H}^2$ is free. Then the orbit space $\mathbb{H}^2/\Gamma$ with the quotient topology admits a hyperbolic structure. Moreover, the associated hyperbolic metric on $\mathbb{H}^2/\Gamma$ is geodesically complete.

Proof. Let $q\colon \mathbb{H}^2 \to \mathbb{H}^2/\Gamma$ be the quotient map. Since Γ acts freely and properly discontinuously, q is a regular covering map. Thus, we can construct an $(\mathbb{H}^2, \mathrm{Isom}^+(\mathbb{H}^2))$-atlas on $\mathbb{H}^2/\Gamma$ as follows:



The domains of the charts φ_i and φ_j are evenly covered open subsets under the regular covering q. The deck transformation $\gamma_z \in \Gamma$, which sends z to the point $\Phi_{ij}(z)$, agrees with ψ_{ij} on a connected neighborhood of z by the unique lifting property.

Observe that the covering map q is a local isometry. The geodesic completeness of $\mathbb{H}^2/\Gamma$ then follows directly from the theorem below. $\square$

145 Theorem. (Ambrose) [Lee2018] Suppose (M, g) and (N, h) are connected Riemannian manifolds with M complete, and $\pi\colon M \to N$ is a local isometry. Then N is complete and π is a smooth covering map.

146 Theorem. [Wol2011] Let Γ_1 and Γ_2 be two Fuchsian groups whose actions on $\mathbb{H}^2$ are free. Then the following are equivalent:

(1) There exists $T \in \mathrm{Isom}^+(\mathbb{H}^2)$ such that $\Gamma_2 = T\Gamma_1 T^{-1}$.

(2) There exists an orientation-preserving isometry $f\colon \mathbb{H}^2/\Gamma_1 \to \mathbb{H}^2/\Gamma_2$.

Proof. Let $q_k\colon \mathbb{H}^2 \to \mathbb{H}^2/\Gamma_k$ be the locally isometric covering maps for $k = 1, 2$.

- (1) $\implies$ (2): Given such a T, define f by $f(q_1(x)) := q_2(T(x))$ for all $x \in \mathbb{H}^2$.
- (2) $\implies$ (1): Given such an f, let $T\colon \mathbb{H}^2 \to \mathbb{H}^2$ be a map such that $q_2 \circ T = f \circ q_1$. Then T is a global isometry by Ambrose's theorem above, and it follows that $\Gamma_2 = T\Gamma_1 T^{-1}$.

§ Riemann Curvature Tensors.

$\boxed{147}$ **Definition.** Let M be an n-dimensional manifold. A smooth section of the vector bundle

$$\mathcal{T}_l^k(M) := \bigsqcup_{p \in M} \hom\left(\bigotimes^k T_p^* M \otimes \bigotimes^l T_p M, \mathbb{R}\right)$$

is called a (k,l)-*tensor*. In any local coordinate system $(U, \varphi = (x^1, \ldots, x^n))$, a (k,l)-tensor s can be expressed as

$$s = \sum_{\substack{1 \leq i_1, \ldots, i_k \leq n \\ 1 \leq j_1, \ldots, j_l \leq n}} A^{i_1,\ldots,i_k}{}_{j_1,\ldots,j_l} \frac{\partial}{\partial x^{i_1}} \otimes \cdots \otimes \frac{\partial}{\partial x^{i_k}} \otimes dx^{j_1} \otimes \cdots \otimes dx^{j_l}$$

for some smooth functions $A^{i_1,\ldots,i_k}{}_{j_1,\ldots,j_l} : U \to \mathbb{R}$, called the *components* of s with respect to (U, φ).

$\boxed{148}$ **Example.** A Riemannian metric on M is a $(0,2)$-tensor.

149 **Theorem.** [Spi1979 2] [Jos2008] For an n-dimensional Riemannian manifold (M, g), there exists a $(1,3)$-tensor called the *Riemann curvature tensor*, denoted R, whose components in a local coordinate system $(U, (x^1, \ldots, x^n))$ are given by:

$$R^i{}_{jkl} := \frac{\partial \Gamma^i_{jl}}{\partial x^k} - \frac{\partial \Gamma^i_{jk}}{\partial x^l} + \sum_{\mu=1}^{n} \left(\Gamma^\mu_{jl} \Gamma^i_{\mu k} - \Gamma^\mu_{jk} \Gamma^i_{\mu l} \right).$$

150 **Theorem.** [Spi1979 2] Let (M, g) be an n-dimensional Riemannian manifold. Then the following are equivalent:

- For every $p \in M$, there is a coordinate chart $(V, \psi = (y^1, \ldots, y^n))$ around p such that g can be expressed in (V, ψ) as

$$g = dy^1 \otimes dy^1 + \cdots + dy^n \otimes dy^n.$$

- The Riemann curvature tensor R is trivial (as a section of the bundle $\mathcal{T}^1_3(M)$).

§ Sectional Curvatures.

Let (M, g) be a Riemannian manifold.

$\boxed{151}$ **Proposition.** [Spi[1979]2] There exists a $(0,4)$-tensor, called the *covariant Riemann curvature tensor* and denoted by $\mathfrak{R}$, whose components in a local coordinate system $(U, (x^1, \ldots, x^n))$ are given by:

$$\mathfrak{R}_{ijkl} := \sum_{\alpha=1}^{n} g_{i\alpha} R^{\alpha}{}_{jkl}.$$

$\boxed{152}$ **Theorem.** [Jos[2008]] Let $p \in M$, and let Π be a 2-dimensional subspace of T_pM with basis $\mathcal{B} = \{v, w\}$. Suppose $(U, (x^1, \ldots, x^n))$ is a local coordinate system with $p \in U$, where $v = \sum_i \xi^i \left.\frac{\partial}{\partial x^i}\right|_p$ and $w = \sum_i \eta^i \left.\frac{\partial}{\partial x^i}\right|_p$. Then the real number K_Π defined by

$$K_\Pi := \frac{\mathfrak{R}_p(v, w, v, w)}{g_p(v,v)g_p(w,w) - g_p(v,w)^2} = \frac{\sum_{i,j,k,l} \mathfrak{R}_{ijkl}\, \xi^i \eta^j \xi^k \eta^l}{\sum_{i,j,k,l} (g_{ik}g_{jl} - g_{ij}g_{kl})\, \xi^i \eta^j \xi^k \eta^l}$$

is independent of the choice of basis $\mathcal{B}$ and is called the *sectional curvature* of the plane Π.

153 Theorem. [Raz2018] [Vil2009] [Küh2002] Let $p \in M$, and let $v, w \in T_pM$ be linearly independent unit tangent vectors. Let $\gamma, \delta \colon (-\varepsilon, \varepsilon) \to M$ be geodesics with $\gamma(0) = \delta(0) = p$ and $\gamma'(0) = v$, $\delta'(0) = w$. Then there exists $\varepsilon' \in (0, \varepsilon)$ such that

$$d^2\left(\gamma(t), \delta(s)\right) = \underbrace{t^2 + s^2 - 2ts\cos\theta}_{\text{Euclidean Part}} + \underbrace{-\frac{1}{3}K_\Pi(\sin^2\theta)t^2s^2}_{\text{Curvature Correction}} + \underbrace{O\left((t^2+s^2)^{5/2}\right)}_{\text{Higher-Order Remainder}}$$

for all $t, s \in (-\varepsilon', \varepsilon')$, where d is the Riemannian distance function, $\theta \in (0, \pi)$ is the angle between v and w (so $\cos\theta = g(v, w)$), and K_Π is the sectional curvature of $\Pi = \operatorname{span}\{v, w\}$.

154 Proposition. If $\dim M = 2$, then for any $p \in M$, the sectional curvature K_p of the unique tangent plane T_pM is given in local coordinates by $K_p = \mathfrak{R}_{1212}(p)/\det(g_{ij}(p))$.

155 Example. The sectional curvature of $\mathbb{H}^n$ (resp. $\mathbb{R}^n$) is -1 (resp. 0) for every tangent plane at every point.

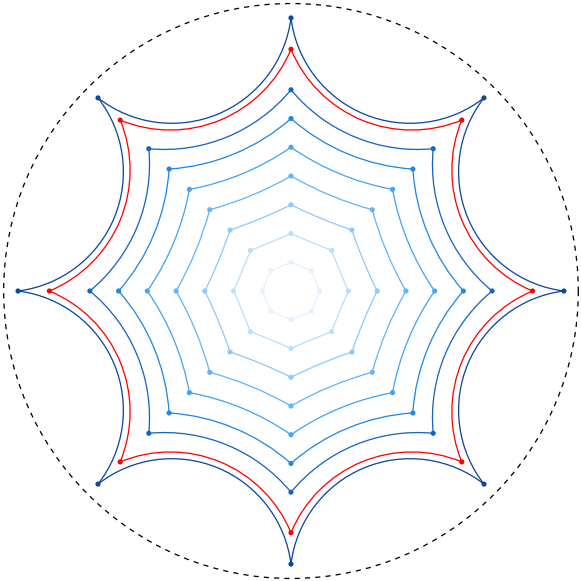
§ Space Forms: The Cartan–Hadamard and Killing–Hopf Theorems.

156 Theorem. Let M be a connected, orientable, complete Riemannian n-manifold without boundary. Denote by $\widetilde{M}$ the universal cover of M. Then we have the following:

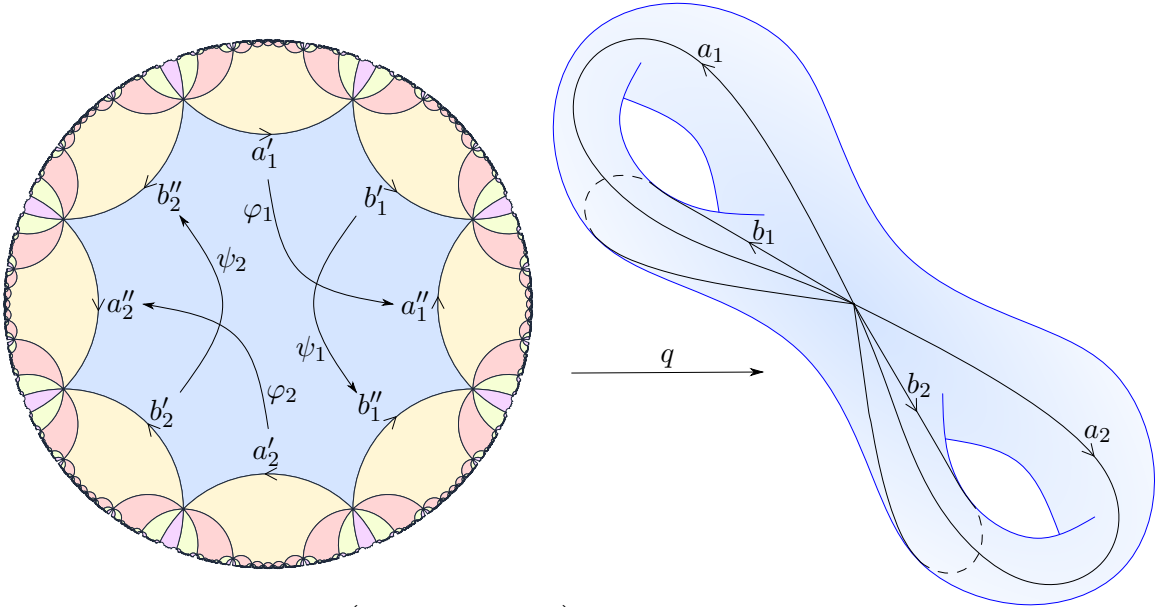
(a) [Man[1881]; Had[1896]; Car[1928]] If all sectional curvatures of M are less than or equal to zero, then $\widetilde{M}$ is diffeomorphic to $\mathbb{R}^n$.

(b) [Kil[1891]; Hop[1926]] If all sectional curvatures of M are equal to -1 for all tangent planes at all points, then
- $\widetilde{M}$ with the pullback metric is isometric to $\mathbb{H}^n$, and
- M is isometric to the orbit space $\mathbb{H}^n/\Gamma$ for some subgroup Γ of $\mathrm{Isom}^+(\mathbb{H}^n)$ which acts freely and properly discontinuously on $\mathbb{H}^n$.

157 Remark. In (b), an analogous result holds when all sectional curvatures of M are equal to 0 (resp. $+1$). In this case, $\widetilde{M}$ is isometric to $\mathbb{R}^n$ (resp. $\mathbb{S}^n$) and M is isometric to $\mathbb{R}^n/\Gamma$ (resp. $\mathbb{S}^n/\Gamma$) for some subgroup Γ of $\mathrm{Isom}^+(\mathbb{R}^n)$ (resp. $\mathrm{Isom}^+(\mathbb{S}^n)$) which acts freely and properly discontinuously on $\mathbb{R}^n$ (resp. $\mathbb{S}^n$).



Fundamental domain with and for S_2, and corresponding $\{8, 8\}$ tessellation Fundamental domain with and for S_2, and corresponding $\{8, 8\}$ tessellation



Fundamental domain with $s = 2\ln\left(\sqrt{2}+1+\sqrt{2+2\sqrt{2}}\right)$ and $\angle_{\text{int}} = 45°$ for S_2, and corresponding $\{8,8\}$ tessellation

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Thank You